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Product Name and Version

KanTime Health v1.0

INTRODUCTION

In this documentation we will list all the steps required to access protected health information based on the documentation of an open API. The API makes health information broadly available using FHIR®, a set of clinical interoperability resources under the umbrella of the HL7 standards organization. FHIR is based on common web standards can be reached through a RESTful protocol in which each FHIR resource has a known URL. This document describes OAuth 2.0 for client applications to authorize, authenticate, and integrate with FHIR-based data systems.

We follow Smart-On-FHIR authentication (OAuth2) and authorization. (Ref: <https://hl7.org/fhir/smart-app-launch/STU2/>)

FHIR Endpoints

Note: Production and Test Server endpoints are subject to change.

Production Endpoint

- FHIR : <https://qafhir.kantimehealth.net:9443/fhirserver/fhir/>
- Authentication Server: <https://qoauth.kantimehealth.net/>

Test Server Endpoint

- FHIR : <https://qafhir.kantimehealth.net:9443/fhirserver/fhir/>
- Authentication Server: <https://qoauth.kantimehealth.net/>

Types of Smart-On-FHIR applications

- Standalone App : SMART on FHIR confidential client with a patient context, refresh token, and OpenID Connect (OIDC) identity token.
- EHR Embedded app: Demonstrate the ability to perform an EHR launch to a SMART on FHIR
- Smart Backend Services App (Multi-patient authorization and API) : These are server-to-server backend applications e.g. Export clinical data for multiple patients in a group. This app is system level app without any UI.

Types of Authentications supported

- Symmetric ("client secret") authentication
- Asymmetric ("private key JWT") authentication
- Public Clients

Steps for Smart App Launch

1. Client App Registration
2. Retrieve .well-known/smart-configuration
3. Obtain authorization code
4. Obtain access token
5. Access FHIR API
6. Refresh Token to obtain new access token

Client APP Registration

Before a SMART app can run against an EHR, the app must be registered with that EHR's authorization service. We are using OAuth 2.0 Client Registration.

Client App Request process

- Third-party application must sign the *API Subscription Agreement* with following details
 - The app name
 - The necessary APIs / Scopes
 - Any redirect URIs
 - Launch_URL (optional)
 - URL to JWK Set (Only for apps supporting asymmetric client authentication).
- The application is created by the EHR admin after confirming the app's registration parameters and communicates a client_id to the app.
 - FHIR APIs (of any supported version) listed within the USCDI v3 core data set will be supported
 - Only reads data from FHIR server

Retrieve .well-known/smart-configuration

In order to obtain launch context and request authorization to access FHIR resources, the app discovers the EHR FHIR server's SMART configuration metadata, including OAuth authorization_endpoint and token_endpoint URLs.

FHIR server makes SMART configuration available from well-known endpoint. You can get Authorization end point and token endpoint.

Request

GET 200 <https://qafhir.kantimehealth.net:9443/fhirserver/fhir/.well-known/smart-configuration>

Response

```
{
  "capabilities": [
    "launch-ehr",
    "launch-standalone",
    "client-public",
    "client-confidential-symmetric",
    "sso-openid-connect",
    "context-banner",
    "context-style",
    "context-ehr-patient",
    "context-ehr-encounter",
    "context-standalone-patient",
    "context-standalone-encounter",
    "permission-offline",
    "permission-patient",
    "permission-user"
  ],
  "revocation_endpoint": "https://qaoauth.kantimehealth.net/outh/token/revoke-token",
  "authorization_endpoint": "https://qaoauth.kantimehealth.net/outh/authorization",
  "token_endpoint": "https://qaoauth.kantimehealth.net/outh/token"
}
```

Error codes

JSON response is returned from the FHIR Server along with the code

200: Ok

401: Unauthorized

400: Bad Request

500: Internal Server Error

Obtain authorization code

The app supplies the following parameters to the EHR’s “authorize” endpoint.

Parameters		
response_type	required	Fixed value: <code>code</code> .
client_id	required	The client’s identifier.
redirect_uri	required	Must match one of the client’s pre-registered redirect URIs.
launch	conditional	When using the <code>EHR Launch</code> flow, this must match the launch value received from the EHR. Omitted when using the <code>Standalone Launch</code> .
scope	required	Must describe the access that the app needs, including scopes like <code>patient/*.rs</code>, <code>openid</code> and <code>fhirUser</code> (if app needs authenticated patient identity) and either: a <code>launch</code> value indicating that the app wants to receive already-established launch context details from the EHR a set of launch context requirements in the form <code>launch/patient</code>, which asks the EHR to establish context on your behalf. Please refer to the scopes supported in the table below
state	required	An opaque value used by the client to maintain state between the request and callback. The authorization server includes this value when redirecting the user-agent back to the client. The parameter SHALL be used for preventing cross-site request forgery or session fixation attacks. The app SHALL use an unpredictable value for the state parameter with at least 122 bits of entropy (e.g., a properly configured random uuid is suitable).
aud	required	URL of the EHR resource server from which the app wishes to retrieve FHIR data. This parameter prevents leaking a genuine bearer token to a counterfeit resource server. (Note that in the case of an <code>EHR launch</code> flow, this <code>aud</code> value is the same as the launch’s <code>iss</code> value.) Note that the <code>aud</code> parameter is semantically equivalent to the <code>resource</code> parameter defined in RFC8707 . SMART’s <code>aud</code> parameter predates RFC8707 and we have decided not to rename it for reasons of backwards compatibility. We might consider renaming SMART’s <code>aud</code> parameter in the future if implementer feedback indicates that alignment would be valuable. For the current release, servers SHALL support the <code>aud</code> parameter and MAY support a <code>resource</code> parameter as a synonym for <code>aud</code> .

Scopes Supported

Scope	Grants
patient/*.read	Permission to read and search any resource for the current patient (see notes on wildcard scopes below).
user/*.read	Permission to read and write all resources that the current user can access (see notes on wildcard scopes below).
openid	Permission to retrieve information about the current logged-in user.
fhirUser	

Scope	Grants
launch	Permission to obtain launch context when app is launched from an EHR.
launch/patient	When launching outside the EHR, ask for a patient to be selected at launch time.
offline_access	Request a refresh_token that can be used to obtain a new access token to replace an expired one, even after the end-user no longer is online after the access token expires.

Here's an example of an authorization request using HTTP GET. You will replace the [redirect_uri], [client_id], [launch_token], [state], [code_challenge], and [audience] placeholders with your own values.

Request

https://ehr/authorize?

```
response_type=code&
client_id=app-client-id&
redirect_uri=https%3A%2F%2Fapp%2Fafter-auth&
launch=xyz123&
scope=launch+patient%2FObservation.rs+patient%2FPatient.rs+openid+fhirUser&
state=98wrghuwoogerg97&
aud=https://ehr/fhir
```

e.g.

https://qaoauth.kantimehealth.net/outh/authorization?response_type=code&client_id=SAMPLE_CONFIDENTIAL_CLIENT_ID&redirect_uri=https%3A%2F%2Finferno.healthit.gov%2Finferno%2Foauth2%2Fstatic%2Fredirect&scope=launch%2Fpatient+openid+fhirUser+offline_access+patient%2FMedication.read+patient%2FAllergyIntolerance.read+patient%2FCarePlan.read+patient%2FCareTeam.read+patient%2FCondition.read+patient%2FDevice.read+patient%2FDiagnosticReport.read+patient%2FDocumentReference.read+patient%2FEncounter.read+patient%2FGoal.read+patient%2FImmunization.read+patient%2FLocation.read+patient%2FMedicationRequest.read+patient%2FObservation.read+patient%2FOrganization.read+patient%2FPatient.read+patient%2FPractitioner.read+patient%2FProcedure.read+patient%2FProvenance.read+patient%2FPractitionerRole.read&state=26a45e37-7445-4e4a-b8fb-24144eccbdc4&aud=https%3A%2F%2Finferno.healthit.gov%2Freference-server%2Ffr4

Response

The EHR's authorization server reviews the request from your application. If approved, the authorization server redirects the browser to the redirect URL supplied in the initial request and appends the following querystring parameter.

- **code**: This parameter contains the authorization code generated by EHR, which will be exchanged for the access token in the next step.
- **state**: The exact value received from the client.

Here is how redirect URL will look like

Location: https://app/after-auth?code=123abc&state=98wrghuwoogerg97

https://inferno.healthit.gov/inferno/oauth2/static/redirect?code=SAMPLE_CODE.bGF1bmNoL3BhdGllbnQgb3BlbmklGZoaXJvc2VyIG9mZmxpbmVfYWVWZjZjZXJ2YXRpb24ucmVhZCBwYXRpZW50L09yZ2FuaXphdGlvbi5yZWZkIHhhdGllbnQvRGV2aWNlLnJlYWQgcGF0aWVudC9EaWFnbm9zdGJlUmVwb3J0LnJlYWQgcGF0aWVudC9Eb2N1bWVudFJlZmVvZW5jZS5yZWZkIHhhdGllbnQvRW5ib3VudGVyLnJlYWQgcGF0aWVudC9Hb2FsLnJlYWQgcGF0aWVudC9JbW11bml6YXRpb24ucmVhZCBwYXRpZW50L0xvY2F0aW9uLnJlYWQgcGF0aWVudC9NZWRpY2F0aW9uUmVxdWVzdC5yZWZkIHhhdGllbnQvT2JzZXJ2YXRpb24ucmVhZCBwYXRpZW50L09yZ2FuaXphdGlvbi5yZWZkIHhhdGllbnQvUGF0aWVudC5yZWZkIHhhdGllbnQvUHJhY3RpdGlvbmVvLnJlYWQgcGF0aWVudC9Qcm9jZWZWR1cmUucmVhZCBwYXRpZW50L1Byb3ZlbnFuY2UucmVhZCBwYXRpZW50L1ByYWN0aXRpb25lclJvbgUucmVhZCA=.ODU=&state=26a45e37-7445-4e4a-b8fb-24144eccbdc4

After receiving the authorization code, your application trades the code for a JSON object containing an access token and contextual information by sending an HTTP POST to the token endpoint using a Content-Type header with value of "application/x-www-form-urlencoded".

Error codes

JSON response is returned from the FHIR Server along with the code

200: OK

401: Unauthorized

400: Bad Request

500: Internal Server Error

Obtain access token

For **public apps**, authentication not required because a client with no secret cannot prove its identity when it issues a call. (The end-to-end system can still be secure because the client comes from a known, https protected endpoint specified and enforced by the redirect uri.)

For **confidential apps**, authentication is required. Confidential clients SHOULD use Asymmetric Authentication if available, and MAY use Symmetric Authentication.

There are two different requests to get access token based on authentication type. In case of Symmetric ("client secret") authentication app issues an HTTP POST to the EHR authorization server's token endpoint URL using content-type application/x-www-form-urlencoded.

In case of Asymmetric authentication app, generate a client authentication assertion and prepare arguments for POST to token API:

Request

Parameters		
grant_type	required	Fixed value: authorization_code for symmetric and client_credentials for assymetric
Code	required	Code that the app received from the authorization server
redirect_uri	required	The same redirect_uri used in the initial authorization request
client_id	conditional	Required for public apps . Omit for confidential apps .
client_assertion_type	conditional	Required for asymmetric authentication. set to urn:ietf:params:oauth:client-assertion-type:jwt-bearer
client_assertion	conditional	Required for asymmetric authentication. set to a JWT signed with your dynamic client's private key
Scope	conditional	system/*.read for backend services type of application (bulk operations)

Symmetric Authentication Request

<https://qaoauth.kantimehealth.net/outh/token>

Payload

grant_type=authorization_code&code=SAMPLE_CODE.bGF1bmNoL3BhdGllbnQgb3BlbmklGZoaXJvc2VylG9mZmxpbmVfYWVWNjZXNzIHdhGllbnQvTWVkaWNhdGlvbi5yZWFKIHdhGllbnQvQWxsZXJneUludG9sZXJhbmNlLnJlYWQgcGF0aWVudC9DYXJlUGxhbi5yZWFKIHdhGllbnQvQ2FyZVRlYW0ucmVhZCBwYXRpZW50L0NvbRpdGlvbi5yZWFKIHdhGllbnQvRGV2aWNlLnJlYWQgcGF0aWVudC9EaWFnbm9zdGljUmVwb3J0LnJlYWQgcGF0aWVudC9Eb2N1bWVudFJlZmVyZW5jZS5yZWFKIHdhGllbnQvRW5jb3VudGVyLnJlYWQgcGF0aWVudC9Hb2FsLnJlYWQgcGF0aWVudC9JbW11bml6YXRpb24ucmVhZCBwYXRpZW50L0xvY2F0aW9uLnJlYWQgcGF0aWVudC9NZWRpY2F0aW9uUmVxdWVzdC5yZWFKIHdhGllbnQvT2JzZXJ2YXRpb24ucmVhZCBwYXRpZW50L09yZ2FuaXphdGlvbi5yZWFKIHdhGllbnQvUGF0aWVudC5yZWFKIHdhGllbnQvUHJhY3RpdGlubmVzLnJlYWQgcGF0aWVudC9Qcm9jZWR1cmUucmVhZCBwYXRpZW50L1Byb3ZlbnFuY2UucmVhZCBwYXRpZW50L1ByYWN0aXRpb25lclJvbGUucmVhZCA%3D.ODU%3D&redirect_uri=https%3A%2F%2Ffinferno.healthit.gov%2Ffinferno%2Foauth2%2Fstatic%2Fredirect

In case of Asymmetric authentication following is a request

```
client_assertion=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiIsInR5cCI6IkpZbzI2ZDFlbn0.eyJpc3MiOiJleUpoYkdjaU9pSkIvekkxTmJlc0luUjVjQ0k2SWtwWFZDSXNjbXRwWkNjNkluSmxaMmx6ZEHkaGRHbHZiaTEwYjJ0bGJpSjkuZXIKcWQydHpYM1ZSYkNjNkltADBkSEET2Hk4eE1DNHhOUzR5TIRjdU56TXZhVzVtWlhKdWJ5OHVklMlZzYkMxcmltOTNiaTlxZDJ0ekxtcHpiMjRpbTENKaFkyTmxjM05VYjJ0bGJpTjZlSEJwY21W5mJpSTZNVFVzSW1saGRDSTZNVFU1TnpReE16RTVOWDAucTR2NE1zYzc0a041MDZLVFowcV9taW55YXBkdzBnd2xUNk1fdWIMNzNTNCIsInN1Yil6lmV5S5mhiR2NpT2IKSVV6STFOaUlzSW5SNWVNDSTZJa3BYVkJkNj0ltdHBaQ0k2SW5KbFoybHpkSEpoZEdsdmJpMTBiMnRsYmlKOS5leUpxZDJ0elgzVnliQ0k2SW1oMGRlQTZMeTh4TUM0eE5TNHlOVEl1TnpNdmFXNW1aWEp1Ynk4dWQyVnNiQzFyYm05M2JpOXFkMnR6TG1wemlyNGlMQ0poWTJObGMzTIViMnRsYm5ORmVlQnBjbVZKYmJlNk1UVXNjWxoZENjNk1UVTV0elF4TXpFNU5YMC5xNHY0TXNjNzRrTjUwNktUWjBxX21pbnlhcEp3MGd3bFQ2TV91aUw3M1M0liwiYXVkljoiaHR0cHM6Ly9pbmZlcm5vLmhlYWx0aGl0Lmdvdidi9yZWZlcmVuY2Utc2VydmlvL29hdXRoL2J1bGstdG9rZW4iLCJleHAiOiJE2NTMxMDY4ODgsIm0aSl6lmZmZjA5YjlxMjhiOWRmNmU3MGU5MWQ2ZDUwMGQxMmFmYmU2YWWE4N2FhMjJzVhMjc5YjZlNTI0YzA0NmM1ZWEIFq.tlx7jlu9wn7QdHHdWYUMf9otY0r7eX62LuVF3kZOSjjw-CP-glqdbDtOKiu-BQ9dodUW4zyk3fs6k48XpAEQLsSXajfEJeDy9uj-WdSmpAHEC0xYHQSzajekdwIDSBRA&client_assertion_type=urn%3Aietf%3Aparams%3Aoauth%3Aclient-assertion-type%3Ajwt-bearer&grant_type=client_credentials&scope=system%2F%2A.read
```

Response

The EHR authorization server SHALL return a JSON object that includes an access token or a message indicating that the authorization request has been denied. The JSON structure includes the following parameters:

Parameters		
access_token	required	The access token issued by the authorization server
token_type	required	Fixed value: <code>Bearer</code>
expires_in	recommended	Lifetime in seconds of the access token, after which the token SHALL NOT be accepted by the resource server
scope	required	Scope of access authorized. Note that this can be different from the scopes requested by the app.
id_token	optional	Authenticated user identity and user details, if requested
refresh_token	optional	Token that can be used to obtain a new access token, using the same or a subset of the original authorization grants

Sample response

```
{
  "access_token": "3e934471-1dbf-4a69-8e1f-227f9fd65428",
  "refresh_token": "31c4476d-6e9d-41a3-a272-065d05c3599d",
  "patient": "85",
  "scope": "launch/patient openid fhirUser offline_access patient/*.*read ",
  "need_patient_banner": false,
  "id_token":
    "eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiIsInR5cCI6IkpZbzI2ZDFlbn0.eyJzdWIiOiJkYjExNzY0MjZlbn0.eyJpc3MiOiJleUpoYkdjaU9pSkIvekkxTmJlc0luUjVjQ0k2SWtwWFZDSXNjbXRwWkNjNkluSmxaMmx6ZEHkaGRHbHZiaTEwYjJ0bGJpSjkuZXIKcWQydHpYM1ZSYkNjNkltADBkSEET2Hk4eE1DNHhOUzR5TIRjdU56TXZhVzVtWlhKdWJ5OHVklMlZzYkMxcmltOTNiaTlxZDJ0ekxtcHpiMjRpbTENKaFkyTmxjM05VYjJ0bGJpTjZlSEJwY21W5mJpSTZNVFVzSW1saGRDSTZNVFU1TnpReE16RTVOWDAucTR2NE1zYzc0a041MDZLVFowcV9taW55YXBkdzBnd2xUNk1fdWIMNzNTNCIsInN1Yil6lmV5S5mhiR2NpT2IKSVV6STFOaUlzSW5SNWVNDSTZJa3BYVkJkNj0ltdHBaQ0k2SW5KbFoybHpkSEpoZEdsdmJpMTBiMnRsYmlKOS5leUpxZDJ0elgzVnliQ0k2SW1oMGRlQTZMeTh4TUM0eE5TNHlOVEl1TnpNdmFXNW1aWEp1Ynk4dWQyVnNiQzFyYm05M2JpOXFkMnR6TG1wemlyNGlMQ0poWTJObGMzTIViMnRsYm5ORmVlQnBjbVZKYmJlNk1UVXNjWxoZENjNk1UVTV0elF4TXpFNU5YMC5xNHY0TXNjNzRrTjUwNktUWjBxX21pbnlhcEp3MGd3bFQ2TV91aUw3M1M0liwiYXVkljoiaHR0cHM6Ly9pbmZlcm5vLmhlYWx0aGl0Lmdvdidi9yZWZlcmVuY2Utc2VydmlvL29hdXRoL2J1bGstdG9rZW4iLCJleHAiOiJE2NTMxMDY4ODgsIm0aSl6lmZmZjA5YjlxMjhiOWRmNmU3MGU5MWQ2ZDUwMGQxMmFmYmU2YWWE4N2FhMjJzVhMjc5YjZlNTI0YzA0NmM1ZWEIFq.tlx7jlu9wn7QdHHdWYUMf9otY0r7eX62LuVF3kZOSjjw-CP-glqdbDtOKiu-BQ9dodUW4zyk3fs6k48XpAEQLsSXajfEJeDy9uj-WdSmpAHEC0xYHQSzajekdwIDSBRA&client_assertion_type=urn%3Aietf%3Aparams%3Aoauth%3Aclient-assertion-type%3Ajwt-bearer&grant_type=client_credentials&scope=system%2F%2A.read",
  "smart_style_url": "https://qaoauth.kantimehealth.net/app/smart-style-url",
  "token_type": "bearer",
  "expires_in": 3600
}
```



```
}
```

Error codes

JSON response is returned from the FHIR Server along with the code

200: Ok

401: Unauthorized

400: Bad Request

500: Internal Server Error

At this point, **the authorization flow is complete.**

Access FHIR API

With a valid access token, the app can access protected EHR data by issuing a FHIR API call to the FHIR endpoint on the EHR's resource server. Only HTTP GET request types are supported while accessing FHIR Resource API. Following resources are supported

Ref: <https://hl7.org/fhir/us/core/STU6.1/profiles-and-extensions.html>

[FHIR Resource](#)

[Allergies and Intolerances](#)

[Assessment and Plan of Treatment](#)

[Care Team Members](#)

[Clinical Notes](#) -US Core DocumentReference Profile&US Core DiagnosticReport Profile for Report and Note Exchange

[Goals](#) - US Core Goal Profile

[Health Concerns](#) - US Core Condition Profile

[Immunizations](#) - US Core Immunization Profile

[Laboratory](#) - US Core Laboratory Result Observation Profile & US Core DiagnosticReport Profile for Laboratory Results Reporting

[Medications](#) - US Core Medication Profile, US Core MedicationRequest Profile, US Core AllergyIntolerance Profile

[Patient Demographics](#) - US Core Patient Profile

[Problems](#) - US Core Condition Profile

[Procedures](#) - US Core Procedure Profile

[Provenance](#) - US Core Provenance Profile

[Smoking Status](#) - US Core Smoking Status Observation Profile

[Unique Device Identifier\(s\) for Patient's Implantable Device\(s\)](#) - US Core ImplantableDevice Profile

[Vital Signs](#)—FHIR Core VitalSigns Profile

[Organization](#) – US Core Organization Profile

[DocumentReference](#) – US Core DocumentReference Profile

[ServiceRequest](#) – US Core ServiceRequest Profile

[Coverage](#) – US Core Coverage Profile

[MedicationDispense](#) – US Core MedicationDispense Profile

[Specimen](#) – US Core Specimen Profile

[RelatedPerson](#) – US Core RelatedPerson Profile

Request : Patient

Search Parameters

_id: GET [Base_URL]/Patient/_search?Patient=[id]

Identifier : GET [Base_URL]/Patient?identifier={system|}[code]

Name: GET [Base_URL]/Patient?name=[string]

birthdate + name :GET [Base_URL]/Patient?birthdate=[date]&name=[string]

gender + name : GET [Base_URL]/Patient?gender={system|}[code]&name=[string]

Response

Code: 200 (Success)

```
{
  "resourceType": "Bundle",
  "id": "6f1d6f2d-c42b-453b-8f5a-8d74c24f6fb6",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://FHIR-Base-URL/api/v4/Patient?_count=10&_id=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b&_page=1"
    }
  ],
  "entry": [
    {
      "id": "17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b",
      "fullUrl": "https://FHIR-Base-URL/api/v4/Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b",
      "resource": {
        "resourceType": "Patient",
        "id": "17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b",
        "meta": {
          "versionId": "3",
          "lastUpdated": "2022-03-24T10:50:49.231Z",
          "profile": [
            "http://hl7.org/fhir/us/core/StructureDefinition/us-core-patient"
          ]
        },
        "text": {
          "status": "generated",
          "div": "<div xmlns='\"http://www.w3.org/1999/xhtml\"'>Generated by <a href='\"https://github.com/synthetichealth/synthea\"'>Synthea</a>.Version identifier: v2.6.1-3-g50f4f58f\\n . Person seed: 4286870567281389426 Population seed: 3</div>"
        },
        "extension": [
          {
            "extension": [
              {
                "url": "ombCategory",
                "valueCoding": {
                  "system": "urn:oid:2.16.840.1.113883.6.238",
                  "code": "2106-3",
                  "display": "White"
                }
              },
              {
                "url": "text",
                "valueString": "White"
              }
            ]
          },
          {
            "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-race"
          },
          {
            "extension": [
              {
                "url": "ombCategory",
                "valueCoding": {
                  "system": "urn:oid:2.16.840.1.113883.6.238",
                  "code": "2186-5",
                  "display": "Non Hispanic or Latino"
                }
              },
              {
                "url": "text",
                "valueString": "Non Hispanic or Latino"
              }
            ]
          },
          {
            "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-ethnicity"
          }
        ]
      }
    }
  ]
}
```

```

    },
    {
      "url": "http://hl7.org/fhir/StructureDefinition/patient-mothersMaidenName",
      "valueString": "Christen366 Murray856"
    },
    {
      "url": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-birthsex",
      "valueCode": "M"
    },
    {
      "url": "http://hl7.org/fhir/StructureDefinition/patient-birthPlace",
      "valueAddress": {
        "city": "Springfield",
        "state": "Massachusetts",
        "country": "US"
      }
    }
  ],
  "identifier": [
    {
      "system": "https://github.com/synthetichealth/synthea",
      "value": "e91975f5-9445-c11f-cabf-c3c6dae161f2"
    },
    {
      "type": {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
            "code": "MR",
            "display": "Medical Record Number"
          }
        ],
        "text": "Medical Record Number"
      },
      "system": "http://hospital.smarthealthit.org",
      "value": "e91975f5-9445-c11f-cabf-c3c6dae161f2"
    },
    {
      "type": {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
            "code": "SS",
            "display": "Social Security Number"
          }
        ],
        "text": "Social Security Number"
      },
      "system": "http://hl7.org/fhir/sid/us-ssn",
      "value": "999-61-9798"
    },
    {
      "type": {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
            "code": "DL",
            "display": "Driver's License"
          }
        ],
        "text": "Driver's License"
      },
      "system": "urn:oid:2.16.840.1.113883.4.3.25",
      "value": "S99940282"
    }
  ],

```

```
"name": [
  {
    "use": "official",
    "family": "Harry",
    "given": [
      "Tom"
    ],
    "prefix": [
      "Mr."
    ],
    "suffix": [
      "Jr."
    ]
  },
  {
    "use": "maiden",
    "given": [
      "Alex"
    ]
  }
],
"telecom": [
  {
    "system": "phone",
    "value": "555-770-2787",
    "use": "home"
  }
],
"gender": "male",
"birthDate": "1941-09-05",
"address": [
  {
    "extension": [
      {
        "extension": [
          {
            "url": "latitude",
            "valueDecimal": 42.1343534042923
          },
          {
            "url": "longitude",
            "valueDecimal": -72.67217549422628
          }
        ]
      },
      {
        "url": "http://hl7.org/fhir/StructureDefinition/geolocation"
      }
    ],
    "line": [
      "599 Schowalter Promenade"
    ],
    "city": "West Springfield",
    "state": "MA",
    "postalCode": "01089",
    "country": "US",
    "period": {
      "start": "1940-09-05"
    }
  }
],
"maritalStatus": {
  "coding": [
    {
      "system": "http://terminology.hl7.org/CodeSystem/v3-MaritalStatus",
      "code": "M",
      "display": "M"
    }
  ]
}
```

```
,
  "text": "M"
},
"multipleBirthBoolean": false,
"communication": [
  {
    "language": {
      "coding": [
        {
          "system": "urn:ietf:bcp:47",
          "code": "en-US",
          "display": "English"
        }
      ],
      "text": "English"
    }
  }
],
"search": {
  "mode": "match",
  "score": 1
}
]
```

Request : AllergyIntolerance

Search Parameters

GET GET [Base_URL]/AllergyIntolerance?Patient=[id]

Response

Code: 200 (Success)

```
{
  "resourceType": "AllergyIntolerance",
  "id": "17e7758c319-7955ecfe-94da-4071-8488-4ef21425ed85",
  "meta": {
    "versionId": "1",
    "lastUpdated": "2022-01-20T11:56:52.889Z",
    "profile": [
      "http://hl7.org/fhir/us/core/StructureDefinition/us-core-allergyintolerance"
    ]
  },
  "clinicalStatus": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/allergyintolerance-clinical",
        "code": "inactive"
      }
    ]
  },
  "verificationStatus": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/allergyintolerance-verification",
        "code": "confirmed"
      }
    ]
  },
  "type": "allergy",
  "category": [
    "food"
  ],
  "criticality": "low",
  "code": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "300916003",
        "display": "Latex allergy"
      }
    ]
  },
  "text": "Latex allergy"
},
"patient": {
  "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b"
},
"recordedDate": "1941-10-14T18:33:18-05:00",
"reaction": [
  {
    "manifestation": [
      {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "271807003",
            "display": "skin rash"
          }
        ]
      }
    ]
  },
  {
    "text": "skin rash"
  }
]
```

```
}  
}  
}  
}
```


Request : CarePlan

GET [Base_URL]/AllergyIntolerance?CarePlan=assess-plan&patient=[id]

e.g.

[https://\[Base_URL\]/CarePlan?category=assess-plan&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b](https://[Base_URL]/CarePlan?category=assess-plan&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b)

Response

Code: 200 (Success)

```
{
  "resourceType": "Bundle",
  "id": "79d7a78e-de9b-4c5c-b51d-45b72963ae1f",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/CarePlan?_count=10&category=assess-plan&patient=Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b&_page=1"
    }
  ],
  "entry": [
    {
      "id": "17e776e13ff-1ea7505d-f475-4f49-adde-30e69df86734",
      "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/CarePlan/17e776e13ff-1ea7505d-f475-4f49-adde-30e69df86734",
      "resource": {
        "resourceType": "CarePlan",
        "id": "17e776e13ff-1ea7505d-f475-4f49-adde-30e69df86734",
        "meta": {
          "versionId": "2",
          "lastUpdated": "2022-01-21T10:56:06.304Z",
          "profile": [
            "http://hl7.org/fhir/us/core/StructureDefinition/us-core-careplan"
          ]
        },
        "text": {
          "status": "generated",
          "div": "<div xmlns='\"http://www.w3.org/1999/xhtml\"'>Care Plan for Fracture care.<br/>Activities: <ul><li>Fracture care</li><li>Fracture care</li></ul><br/>Care plan is meant to treat Fracture of clavicle.</div>"
        },
        "status": "active",
        "intent": "order",
        "category": [
          {
            "coding": [
              {
                "system": "http://hl7.org/fhir/us/core/CodeSystem/careplan-category",
                "code": "assess-plan"
              }
            ]
          }
        ],
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "384758001",
            "display": "Self-care interventions (procedure)"
          }
        ],
        "text": "Self-care interventions (procedure)"
      }
    ],
    {
      "subject": {
```

```
"reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b"
},
"encounter": {
  "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
},
"period": {
  "start": "1941-10-02T18:33:18-05:00"
},
"careTeam": [
  {
    "reference": "CareTeam/17e775e7546-36e6a4cd-264e-4a94-92fa-cdf1b2fa30c1"
  }
],
"activity": [
  {
    "detail": {
      "code": {
        "coding": [
          {
            "system": "http://snomed.info/sct",
            "code": "409002",
            "display": "Food allergy diet"
          }
        ]
      },
      "text": "Food allergy diet"
    },
    "status": "in-progress",
    "location": {
      "reference": "Location/17e7750e68f-6266aec8-0cd3-443e-adb9-be37f3bf25f2",
      "display": "HOLYOKE MEDICAL CENTER"
    }
  }
],
{
  "detail": {
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "58332002",
          "display": "Allergy education"
        }
      ]
    },
    "text": "Allergy education"
  },
  "status": "in-progress",
  "location": {
    "reference": "Location/17e7750e68f-6266aec8-0cd3-443e-adb9-be37f3bf25f2",
    "display": "HOLYOKE MEDICAL CENTER"
  }
},
{
  "detail": {
    "code": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "58332002",
          "display": "Allergy education"
        }
      ]
    },
    "text": "Allergy education"
  },
  "status": "in-progress",
  "location": {
```

```
      "reference": "Location/17e7750e68f-6266aec8-0cd3-443e-adb9-be37f3bf25f2",  
      "display": "HOLYOKE MEDICAL CENTER"  
    }  
  }  
}  
],  
"search": {  
  "mode": "match",  
  "score": 1  
}  
}  
]
```

Request : CareTeam

GET [Base_URL]/CareTeam?patient=[id]&status=active

e.g.

https://[Base_URL]/CareTeam?patient=17e77411359-8837d878-ea1e-45b5-bb89-882dfdb2be4b&status=active
Response

Code: 200 (Success)

```
{
  "resourceType": "Bundle",
  "id": "492509fd-0650-4e2e-834f-41fc7b132a5e",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/CareTeam?_count=10&patient=17e77411359-8837d878-ea1e-45b5-bb89-882dfdb2be4b&status=http://hl7.org/fhir/care-team-status%7Cactive&_page=1"
    }
  ],
  "entry": [
    {
      "id": "17e775e7546-36e6a4cd-264e-4a94-92fa-cdf1b2fa30c1",
      "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/CareTeam/17e775e7546-36e6a4cd-264e-4a94-92fa-cdf1b2fa30c1",
      "resource": {
        "resourceType": "CareTeam",
        "id": "17e775e7546-36e6a4cd-264e-4a94-92fa-cdf1b2fa30c1",
        "meta": {
          "versionId": "1",
          "lastUpdated": "2022-01-20T12:03:06.182Z",
          "profile": [
            "http://hl7.org/fhir/us/core/StructureDefinition/us-core-careteam"
          ]
        },
        "status": "active",
        "subject": {
          "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882dfdb2be4b"
        },
        "encounter": {
          "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
        },
        "period": {
          "start": "1941-10-02T18:33:18-05:00"
        },
        "participant": [
          {
            "role": [
              {
                "coding": [
                  {
                    "system": "http://snomed.info/sct",
                    "code": "116154003",
                    "display": "Patient"
                  }
                ]
              },
              {
                "text": "Patient"
              }
            ]
          }
        ],
        "member": {
          "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882dfdb2be4b",
          "display": "Mr. Dustin31 Ritchie586"
        }
      }
    }
  ]
}
```

```
}
},
{
  "role": [
    {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "223366009",
          "display": "Healthcare professional (occupation)"
        }
      ],
      "text": "Healthcare professional (occupation)"
    }
  ],
  "member": {
    "reference": "Practitioner/17e774bb850-6b7b7f50-ed52-4523-a3a8-60fb0ef71c5b",
    "display": "Dr. Patricia625 Kilback373"
  }
},
{
  "role": [
    {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "224891009",
          "display": "Healthcare services (qualifier value)"
        }
      ],
      "text": "Healthcare services (qualifier value)"
    }
  ],
  "member": {
    "reference": "Organization/17e774dca91-e7e4e9e3-390e-4c10-83ae-88beabc176e2",
    "display": "HOLYOKE MEDICAL CENTER"
  }
},
{
  "managingOrganization": [
    {
      "reference": "Organization/17e774dca91-e7e4e9e3-390e-4c10-83ae-88beabc176e2",
      "display": "HOLYOKE MEDICAL CENTER"
    }
  ]
},
{
  "search": {
    "mode": "match",
    "score": 1
  }
}
]
```

Request: Problems / Health Concern

Search Parameters

```
GET [base]/Condition?patient=[id]&category={system | }[code]
GET [base]/Condition?patient=[id]&category=http://terminology.hl7.org/CodeSystem/condition-category|problem-list-item
GET [base]/Condition?patient=[id]&category=http://hl7.org/fhir/us/core/CodeSystem/condition-category|health-concern
GET [base]/Condition?patient=[id]&category=http://terminology.hl7.org/CodeSystem/condition-category|encounter-diagnosis
e.g
[Base_URL]/Condition?patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b&category=http://hl7.org/fhir/us/core/CodeSystem/condition-category|health-concern
```

Response

Code: 200 (Success)

```
{
  "resourceType": "Bundle",
  "id": "f537c30e-fab7-45f0-8167-4ab7a48386ec",
  "type": "searchset",
  "total": 2,
  "link": [
    {
      "relation": "self",
      "url": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Condition?_count=10&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b&_page=1"
    }
  ],
  "entry": [
    {
      "id": "17e7777ce8c-c66752fb-ee1b-40c2-ba5a-2e6ee89b3e1a",
      "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Condition/17e7777ce8c-c66752fb-ee1b-40c2-ba5a-2e6ee89b3e1a",
      "resource": {
        "resourceType": "Condition",
        "id": "17e7777ce8c-c66752fb-ee1b-40c2-ba5a-2e6ee89b3e1a",
        "meta": {
          "versionId": "1",
          "lastUpdated": "2022-01-20T12:30:47.436Z",
          "profile": [
            "http://hl7.org/fhir/us/core/StructureDefinition/us-core-condition"
          ]
        },
        "clinicalStatus": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/condition-clinical",
              "code": "resolved"
            }
          ]
        },
        "verificationStatus": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/condition-ver-status",
              "code": "confirmed"
            }
          ]
        },
        "category": [
          {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/condition-category",
                "code": "encounter-diagnosis",
                "display": "Encounter Diagnosis"
              }
            ]
          }
        ]
      }
    }
  ]
}
```

```

    }
  ]
},
"code": {
  "coding": [
    {
      "system": "http://snomed.info/sct",
      "code": "444814009",
      "display": "Viral sinusitis (disorder)"
    }
  ],
  "text": "Viral sinusitis (disorder)"
},
"subject": {
  "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882dfb2be4b"
},
"encounter": {
  "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
},
"onsetDateTime": "1940-11-09T18:33:18-05:00",
"abatementDateTime": "1940-11-30T18:33:18-05:00",
"recordedDate": "1940-11-09T18:33:18-05:00"
},
"search": {
  "mode": "match",
  "score": 1
}
},
{
  "id": "17eb029f925-40603327-0ac3-48dd-972b-6c882b8751f5",
  "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Condition/17eb029f925-40603327-0ac3-48dd-972b-6c882b8751f5",
  "resource": {
    "resourceType": "Condition",
    "id": "17eb029f925-40603327-0ac3-48dd-972b-6c882b8751f5",
    "meta": {
      "versionId": "1",
      "lastUpdated": "2022-01-31T12:44:07.847Z",
      "profile": [
        "http://hl7.org/fhir/us/core/StructureDefinition/us-core-condition"
      ]
    },
    "clinicalStatus": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/condition-clinical",
          "code": "resolved"
        }
      ]
    },
    "verificationStatus": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/condition-ver-status",
          "code": "confirmed"
        }
      ]
    },
    "category": [
      {
        "extension": [
          {
            "url": "http://hl7.org/fhir/StructureDefinition/data-absent-reason",
            "valueCode": "unknown"
          }
        ]
      ]
    ]
  }
}

```

```
}
],
"code": {
  "coding": [
    {
      "system": "http://snomed.info/sct",
      "code": "195662009",
      "display": "Acute viral pharyngitis (disorder)"
    }
  ],
  "text": "Acute viral pharyngitis (disorder)"
},
"subject": {
  "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdfb2be4b"
},
"encounter": {
  "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
},
"onsetDateTime": "1961-06-25T01:11:45-04:00",
"abatementDateTime": "1961-07-07T01:11:45-04:00",
"recordedDate": "1961-06-25T01:11:45-04:00"
},
"search": {
  "mode": "match",
  "score": 1
}
}
]
```


Request: Implantable Device

Search Parameters

GET [Base_URL]/Device?patient=[id]

e.g. [Base_URL]/Device?patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b

Response

Code: 200 (Success)

```
{
  "resourceType": "Bundle",
  "id": "5804551e-120d-4164-b05b-9705fad44fa5",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Device?_count=10&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b&_page=1"
    }
  ],
  "entry": [
    {
      "id": "17e777b3f34-04bc3627-92d3-43bb-92e7-35533110f750",
      "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Device/17e777b3f34-04bc3627-92d3-43bb-92e7-35533110f750",
      "resource": {
        "resourceType": "Device",
        "id": "17e777b3f34-04bc3627-92d3-43bb-92e7-35533110f750",
        "meta": {
          "versionId": "1",
          "lastUpdated": "2022-01-20T12:34:32.884Z",
          "profile": [
            "http://hl7.org/fhir/us/core/StructureDefinition/us-core-implantable-device"
          ]
        },
        "udiCarrier": [
          {
            "deviceIdentifier": "43069338026389",
            "carrierHRF": "(01)43069338026389(11)000302(17)250317(10)1134(21)842026117977"
          }
        ],
        "status": "active",
        "distinctIdentifier": "43069338026389",
        "manufactureDate": "2000-03-02T18:33:18-05:00",
        "expirationDate": "2025-03-17T19:33:18-04:00",
        "lotNumber": "1134",
        "serialNumber": "842026117977",
        "deviceName": [
          {
            "name": "Implantable defibrillator, device (physical object)",
            "type": "user-friendly-name"
          }
        ],
        "type": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "72506001",
              "display": "Implantable defibrillator, device (physical object)"
            }
          ]
        },
        "text": "Implantable defibrillator, device (physical object)"
      }
    }
  ],
}
```

```
"patient": {  
  "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b"  
}  
},  
"search": {  
  "mode": "match",  
  "score": 1  
}  
}  
]
```

Request: Diagnostic Report/ Clinical Notes/ DocumentReference

Search Parameters

patient : GET [Base_URL]/DiagnosticReport?patient=[id]
patient + code : GET [Base_URL]/DiagnosticReport?patient=[id]&code={system|}[code]
patient + category: GET [Base_URL]/DiagnosticReport?patient=[id]&category=LAB
patient : GET [Base_URL]/ DocumentReference?patient=[id]

e.g.

[https://\[Base_URL\]/DiagnosticReport?category=LP29684-5&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b](https://[Base_URL]/DiagnosticReport?category=LP29684-5&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b)
[https://\[Base_URL\]/DiagnosticReport?category=LP7839-6&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b](https://[Base_URL]/DiagnosticReport?category=LP7839-6&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b)
[https://\[Base_URL\]/DiagnosticReport?category=LP29708-2&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b](https://[Base_URL]/DiagnosticReport?category=LP29708-2&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b)
[https://\[Base_URL\]/DocumentReference?patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b](https://[Base_URL]/DocumentReference?patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b)

Response

Code: 200 (Success)

```
{
  "resourceType": "Bundle",
  "id": "a87a8318-2dfc-488d-9f0a-6b07d7c29271",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/DiagnosticReport?_count=10&category=LP29684-5&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b&_page=1"
    }
  ],
  "entry": [
    {
      "id": "17e7b6e6f1c-119e9b8b-11b9-448b-b760-dc116013b654",
      "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/DiagnosticReport/17e7b6e6f1c-119e9b8b-11b9-448b-b760-dc116013b654",
      "resource": {
        "resourceType": "DiagnosticReport",
        "id": "17e7b6e6f1c-119e9b8b-11b9-448b-b760-dc116013b654",
        "meta": {
          "versionId": "1",
          "lastUpdated": "2022-01-21T06:59:02.053Z",
          "profile": [
            "http://hl7.org/fhir/us/core/StructureDefinition/us-core-diagnosticreport-note"
          ]
        },
        "status": "final",
        "category": [
          {
            "coding": [
              {
                "system": "http://loinc.org",
                "code": "LP29684-5",
                "display": "Radiology"
              }
            ],
            "text": "Radiology"
          }
        ],
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "34117-2",
            }
          ]
        }
      }
    }
  ]
}
```

```
    "display": "History and physical note"
  }
},
"text": "History and physical note"
},
"subject": {
  "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b"
},
"encounter": {
  "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
},
"effectiveDateTime": "1974-01-24T19:33:18-04:00",
"issued": "1974-01-24T19:33:18.715-04:00",
"performer": [
  {
    "reference": "Practitioner/17e774bb850-6b7b7f50-ed52-4523-a3a8-60fb0ef71c5b",
    "display": "Dr. Blossom971 Christiansen251"
  },
  {
    "reference": "Practitioner/17e774bb850-6b7b7f50-ed52-4523-a3a8-60fb0ef71c5b"
  },
  {
    "reference": "Organization/17e774dca91-e7e4e9e3-390e-4c10-83ae-88beabc176e2"
  }
],
"presentedForm": [
  {
    "contentType": "text/plain",
    "data":
"CjE5NzQtMDEtMjQKCiMgQ2hpZWYgQ29tcGxhaW50Ci0gRGVjcmVhc2VklGlulEp1ZGdlbWVudAotIENvbmZ1c2lrbgotIE5hc2FsiENvbmdlc3Rpb24KLSBTbmVlemluZyBGaXRzCi0gTmFzYWwgRGZlY2hhcmdlCi0gQ291Z2gKCgojIEhpc3Rvcnkgb2YgUHJlc2VudCBJbGxuZXNzCkR1c3RpbjMxIGlzlGEgMzMeWVhcn1vbGQgbm9uLWVhpc3BhbmJlIHdoaXRlIG1hbGUuIFBhdGllbnQgaGFzIGEgaGlzdG9yeSBvZiBhY3V0ZSBhbGxlcmdpYyByZWJjdGlwbWgYWn1dGUgYnYvbmNoaXRpcyAoZGlzb3JkZXIpLCBsYWNIcmF0aW9uIG9mIHROaWdoLCBvdGl0aXMgbWVkaWEsiH2pcmFsiHNpbnVzaXRpcyAoZGlzb3JkZXIpLCBzcHJhaW4gb2YgYW5rbGUsIGZpcnN0IGRlZ3JlZSBidXJuLCBmcmlFjdHVyZSBvZiBhbmRlZSBwY2hpYGRob29kIGFzdGhtYS4KCiMgU29jaWFsiEhpc3RvcnkKUGF0aWVudCBpcyBtYXJyaWVvLiBQYXRpZW50IGlzlHRIbXBvcmlFyaWx5IGhvbWVvZXNzLiBQYXRpZW50IGlzlGFuIGFjdGJlZSBzbW9rZXIgaW5kIGlzlGFuIGFsY29ob2xpYy4gUGF0aWVudCBpZGVudGimaWVzIGFzIGhldGVyb3NieHVhbC4KCIbhdGllbnQgY29tZXMgaW50IGlzlGFuIGFjdGJlZSBzbW9rSBzb2Npb2Vjb25vbWljIGJhY2tncm91bmQuIFBhdGllbnQgZGlIG5vdCBmaW5pc2ggaGlnaCBzY2hvb2wulFBhdGllbnQgY3VyYmVudGx5IGhhcyBBZXRUYS4KCiMgQWxsZXJnaWVzCmxdGV4IGFsbGVyZ3k5IGFsbGVyZ3kgdG8gc295YQoKlyBNZWRpY2F0aW9ucwphY2V0YW1pbm9waGVudMyNSBtZyBvcmlHRhYmxdDsgMTIwIGFjdHVhdCBmbHV0aWNhc29uZSBwcm9waW9uYXRlIDAuMDQ0IG1nL2FjdHVhdCBtZXRIcmVklIGRvc2UgaW5oYWxlcjsgbmlhMDIwNTAzIDwMCBhY3R1YXQgYWxidXRlcm9sIDAuMDkgbWcvYWN0dWF0IG1ldGVyZWQgZG9zZSBpbmhbbGVyOyBwcmVkbmlzb25lIDUgbWcgb3JhbCB0YWJsZXQ7IGFzcGlyaW4gODEgbWcgb3JhbCB0YWJsZXQKCiMgQXNzZXNzbWVudCBhbmQgUGxhbgoKCiMjIFBsYW4KUGF0aWVudCB3YXMGZ2l2ZW4gdGhIGZvbGxvd2luZyBpbW11bml6YXRpb25zOiBpbmZsdWVuemEsiHNIYXNvbmFsLCBpbmplY3RhYmxiLCBwcmVzZXJ2YXRpdmlUgZnJlZS4gCiRoZSBmb2xsb3dpbmcmcHJvY2VkdXJlcyB3ZXJlIGNvbmR1Y3RlZDoKLSBtZWRpY2F0aW9uIHJlY29uY2lsaWFOaW9uIChwcm9jZWR1cmUpCiRoZSBwYXRpZW50IHdhcyBwcmVzY3JpYmVklHROZSBmb2xsb3dpbmcmcgbWVkaWNhdGlvbnM6Ci0gMTIwIGFjdHVhdCBmbHV0aWNhc29uZSBwcm9waW9uYXRlIDAuMDQ0IG1nL2FjdHVhdCBtZXRIcmVklIGRvc2UgaW5oYWxlcgotIG5kYTAyMDUwMyAyMDAgYWN0dWF0IGFsYnV0ZXJvbcAwLjA5IG1nL2FjdHVhdCBtZXRIcmVklIGRvc2UgaW5oYWxlcgo="
  }
]
},
"search": {
  "mode": "match",
  "score": 1
}
}
}
```

Request: Laboratory Result Observation

Search Parameters

patient : GET [Base_URL]/DiagnosticReport?patient=[id]
patient + code : GET [Base_URL]/DiagnosticReport?patient=[id]&code={system|}[code]
patient + category: GET [Base_URL]/DiagnosticReport?patient=[id]&category=LAB
e.g.

[https://\[Base_URL\]/DiagnosticReport?category=http://terminology.hl7.org/CodeSystem/v2-0074%7CLAB&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b](https://[Base_URL]/DiagnosticReport?category=http://terminology.hl7.org/CodeSystem/v2-0074%7CLAB&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b)
[https://\[Base_URL\]/DiagnosticReport?category=LAB&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b](https://[Base_URL]/DiagnosticReport?category=LAB&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b)
[https://\[Base_URL\]/DiagnosticReport?category=LAB&patient=Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b](https://[Base_URL]/DiagnosticReport?category=LAB&patient=Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b)

Response

Code: 200 (Success)

```
{
  "resourceType": "Bundle",
  "id": "019290b1-c4e1-4025-862c-60ffc3a6160b",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/DiagnosticReport?_count=10&category=http://terminology.hl7.org/CodeSystem/v2-0074%7CLAB&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b&_page=1"
    }
  ],
  "entry": [
    {
      "id": "17e7b7b7e66-36f4cf84-3482-4e36-8cca-13cc832b3bed",
      "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/DiagnosticReport/17e7b7b7e66-36f4cf84-3482-4e36-8cca-13cc832b3bed",
      "resource": {
        "resourceType": "DiagnosticReport",
        "id": "17e7b7b7e66-36f4cf84-3482-4e36-8cca-13cc832b3bed",
        "meta": {
          "versionId": "1",
          "lastUpdated": "2022-01-21T07:13:17.926Z",
          "profile": [
            "http://hl7.org/fhir/us/core/StructureDefinition/us-core-diagnosticreport-lab"
          ]
        },
        "status": "final",
        "category": [
          {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/v2-0074",
                "code": "LAB",
                "display": "Laboratory"
              }
            ]
          }
        ],
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "51990-0",
              "display": "Basic Metabolic Panel"
            }
          ]
        }
      }
    }
  ],
}
```

```
    "text": "Basic Metabolic Panel"
  },
  "subject": {
    "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b"
  },
  "encounter": {
    "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
  },
  "effectiveDateTime": "2017-09-28T19:33:18-04:00",
  "issued": "2017-09-28T19:33:18.715-04:00",
  "performer": [
    {
      "reference": "Organization/17e774dca91-e7e4e9e3-390e-4c10-83ae-88beabc176e2",
      "display": "PCP170967"
    },
    {
      "reference": "Practitioner/17e774bb850-6b7b7f50-ed52-4523-a3a8-60fb0ef71c5b"
    },
    {
      "reference": "Organization/17e774dca91-e7e4e9e3-390e-4c10-83ae-88beabc176e2"
    }
  ],
  "result": [
    {
      "reference": "Observation/17e77809e90-d5cfd26d-ab5d-4f36-bc32-1ee799b2c278",
      "display": "Glucose"
    },
    {
      "reference": "Observation/17e7785ea69-6c9f88d9-0553-4216-be17-f908ebe0b371",
      "display": "Urea Nitrogen"
    }
  ]
},
"search": {
  "mode": "match",
  "score": 1
}
}
```

Request: Goal

GET https://[Base_URL]/Goal?patient=[id]

e.g. https://[Base_URL]/Goal?patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b

Response

Code: 200 (Success)

```
{
  "resourceType": "Bundle",
  "id": "6f2db230-b4f4-4447-856b-3f398377a95f",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Goal?_count=10&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b&_page=1"
    }
  ],
  "entry": [
    {
      "id": "17e7b8b62f8-a4f77df6-b97e-4ece-82ff-36480c9cc38c",
      "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Goal/17e7b8b62f8-a4f77df6-b97e-4ece-82ff-36480c9cc38c",
      "resource": {
        "resourceType": "Goal",
        "id": "17e7b8b62f8-a4f77df6-b97e-4ece-82ff-36480c9cc38c",
        "meta": {
          "versionId": "1",
          "lastUpdated": "2022-01-21T07:30:39.48Z",
          "profile": [
            "http://hl7.org/fhir/us/core/StructureDefinition/us-core-goal"
          ]
        },
        "lifecycleStatus": "active",
        "description": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "281004",
              "display": "Alcoholic dementia"
            }
          ],
          "text": "Alcoholic dementia"
        },
        "subject": {
          "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b"
        },
        "target": [
          {
            "dueDate": "2020-12-07"
          }
        ]
      },
      "search": {
        "mode": "match",
        "score": 1
      }
    }
  ]
}
```

Request: Immunization

GET https://[Base_URL]/Immunization?patient=[id]

e.g. https://[Base_URL]/Immunization?patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b

Response

Code: 200 (Success)

```
{
  "resourceType": "Bundle",
  "id": "206f31dc-9636-4e6f-a652-40c307eebe40",
  "type": "searchset",
  "total": 2,
  "link": [
    {
      "relation": "self",
      "url": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Immunization?_count=10&patient=Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b&_page=1"
    }
  ],
  "entry": [
    {
      "id": "17e7b8dc9ab-8abc85e3-0ab1-470f-ae8-1f4e9a3597bb",
      "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Immunization/17e7b8dc9ab-8abc85e3-0ab1-470f-ae8-1f4e9a3597bb",
      "resource": {
        "resourceType": "Immunization",
        "id": "17e7b8dc9ab-8abc85e3-0ab1-470f-ae8-1f4e9a3597bb",
        "meta": {
          "versionId": "1",
          "lastUpdated": "2022-01-21T07:33:16.843Z",
          "profile": [
            "http://hl7.org/fhir/us/core/StructureDefinition/us-core-immunization"
          ]
        },
        "status": "not-done",
        "statusReason": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/v3-ActReason",
              "code": "OSTOCK",
              "display": "product out of stock"
            }
          ],
          "text": "product out of stock"
        },
        "vaccineCode": {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/data-absent-reason",
              "code": "unknown",
              "display": "Unknown"
            }
          ],
          "text": "Unknown"
        },
        "patient": {
          "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b"
        },
        "encounter": {
          "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
        },
        "occurrenceString": "1946-08-29T19:33:18-04:00",
        "primarySource": true,
        "location": {
          "reference": "Location/17e7750e68f-6266aec8-0cd3-443e-adb9-be37f3bf25f2",
          "display": "RIVER BEND MEDICAL GROUP - CHICOPEE OFFICE - URGENT CARE"
        }
      }
    }
  ]
}
```



```

    }
  },
  "search": {
    "mode": "match",
    "score": 1
  }
},
{
  "id": "17eb02c0f47-3b8a6c05-ae47-4002-9001-fd61626bfca5",
  "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Immunization/17eb02c0f47-3b8a6c05-ae47-4002-9001-fd61626bfca5",
  "resource": {
    "resourceType": "Immunization",
    "id": "17eb02c0f47-3b8a6c05-ae47-4002-9001-fd61626bfca5",
    "meta": {
      "versionId": "1",
      "lastUpdated": "2022-01-31T12:46:24.583Z",
      "profile": [
        "http://hl7.org/fhir/us/core/StructureDefinition/us-core-immunization"
      ]
    },
    "status": "not-done",
    "statusReason": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/v3-ActReason",
          "code": "OSTOCK",
          "display": "product out of stock"
        }
      ]
    },
    "text": "product out of stock"
  },
  "vaccineCode": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/data-absent-reason",
        "code": "unknown",
        "display": "Unknown"
      }
    ]
  },
  "text": "Unknown"
},
  "patient": {
    "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882dfb2be4b"
  },
  "encounter": {
    "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
  },
  "occurrenceString": "1946-08-29T19:33:18-04:00",
  "primarySource": true,
  "location": {
    "reference": "Location/17e7750e68f-6266aec8-0cd3-443e-adb9-be37f3bf25f2",
    "display": "RIVER BEND MEDICAL GROUP - CHICOPEE OFFICE - URGENT CARE"
  }
},
  "search": {
    "mode": "match",
    "score": 1
  }
}
]
}

```

Request: Medication

GET [https://\[Base_URL\]/MedicationRequest?patient=\[id\]](https://[Base_URL]/MedicationRequest?patient=[id])

e.g. [https://\[Base_URL\]/MedicationRequest?patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b&intent=order&status=active](https://[Base_URL]/MedicationRequest?patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b&intent=order&status=active)

Response

Code: 200 (Success)

```
{
  "resourceType": "MedicationRequest",
  "id": "17e7b9b263e-9cd8f64f-f7e1-447e-8583-a753342d13c6",
  "meta": {
    "versionId": "1",
    "lastUpdated": "2022-01-21T07:47:52.51Z",
    "profile": [
      "http://hl7.org/fhir/us/core/StructureDefinition/us-core-medicationrequest"
    ]
  },
  "status": "stopped",
  "intent": "plan",
  "reportedBoolean": true,
  "medicationCodeableConcept": {
    "coding": [
      {
        "system": "http://www.nlm.nih.gov/research/umls/rxnorm",
        "code": "2123111",
        "display": "NDA020503 200 ACTUAT Albuterol 0.09 MG/ACTUAT Metered Dose Inhaler"
      }
    ],
    "text": "NDA020503 200 ACTUAT Albuterol 0.09 MG/ACTUAT Metered Dose Inhaler"
  },
  "subject": {
    "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b"
  },
  "authoredOn": "2020-08-13T19:33:18-04:00",
  "requester": {
    "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b"
  },
  "reasonReference": [
    {
      "reference": "Condition/17e7777ce8c-c66752fb-ee1b-40c2-ba5a-2e6ee89b3e1a"
    }
  ],
  "dosageInstruction": [
    {
      "sequence": 1,
      "text": "Take as needed.",
      "asNeededBoolean": true
    }
  ]
}
```

Request: Smoking Status

GET [https://\[Base_URL\]/Observation?code=72166-2&patient=\[id\]](https://[Base_URL]/Observation?code=72166-2&patient=[id])

e.g. [https://\[Base_URL\]/Observation?code=72166-2&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b](https://[Base_URL]/Observation?code=72166-2&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b)

Response

Code: 200 (Success)

```
{
  "resourceType": "Bundle",
  "id": "2ea85043-b450-40f1-8d2d-751af1c10abb",
  "type": "searchset",
```

```

"total": 2,
"link": [
  {
    "relation": "self",
    "url": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Observation?_count=10&code=72166-2&patient=17e77411359-8837d878-
ea1e-45b5-bb89-882fdb2be4b&_page=1"
  }
],
"entry": [
  {
    "id": "17e77809e90-d5cfd26d-ab5d-4f36-bc32-1ee799b2c278",
    "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Observation/17e77809e90-d5cfd26d-ab5d-4f36-bc32-1ee799b2c278",
    "resource": {
      "resourceType": "Observation",
      "id": "17e77809e90-d5cfd26d-ab5d-4f36-bc32-1ee799b2c278",
      "meta": {
        "versionId": "1",
        "lastUpdated": "2022-01-20T12:40:24.976Z",
        "profile": [
          "http://hl7.org/fhir/us/core/StructureDefinition/us-core-smokingstatus"
        ]
      },
      "status": "final",
      "category": [
        {
          "coding": [
            {
              "system": "http://terminology.hl7.org/CodeSystem/observation-category",
              "code": "survey",
              "display": "survey"
            }
          ]
        }
      ],
      "code": {
        "coding": [
          {
            "system": "http://loinc.org",
            "code": "72166-2",
            "display": "Tobacco smoking status NHIS"
          }
        ]
      },
      "text": "Tobacco smoking status NHIS"
    },
    "subject": {
      "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b"
    },
    "encounter": {
      "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
    },
    "effectivePeriod": {
      "start": "1940-09-05T19:33:18-04:00",
      "end": "1940-09-05T20:33:18-04:00"
    },
    "issued": "1940-09-05T19:33:18.715-04:00",
    "valueCodeableConcept": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "266919005",
          "display": "Never smoker"
        }
      ]
    },
    "text": "Never smoker"
  }
],

```

```

"search": {
  "mode": "match",
  "score": 1
},
{
  "id": "17e7785ea69-6c9f88d9-0553-4216-be17-f908ebe0b371",
  "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Observation/17e7785ea69-6c9f88d9-0553-4216-be17-f908ebe0b371",
  "resource": {
    "resourceType": "Observation",
    "id": "17e7785ea69-6c9f88d9-0553-4216-be17-f908ebe0b371",
    "meta": {
      "versionId": "1",
      "lastUpdated": "2022-01-20T12:46:12.073Z",
      "profile": [
        "http://hl7.org/fhir/us/core/StructureDefinition/us-core-smokingstatus"
      ]
    },
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "survey",
            "display": "survey"
          }
        ]
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "72166-2",
          "display": "Tobacco smoking status NHIS"
        }
      ]
    },
    "text": "Tobacco smoking status NHIS"
  },
  "subject": {
    "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b"
  },
  "encounter": {
    "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
  },
  "effectivePeriod": {
    "start": "1940-09-05T19:33:18-04:00",
    "end": "1940-09-05T20:33:18-04:00"
  },
  "issued": "1940-09-05T19:33:18.715-04:00",
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "266919005",
        "display": "Never smoker"
      }
    ]
  },
  "text": "Never smoker"
},
"search": {
  "mode": "match",
  "score": 1
}

```

}
1
}

Request: Procedure

GET https://[Base_URL]/Procedure?patient=[id]

e.g. https://[Base_URL]/Procedure?patient=17e77411359-8837d878-ea1e-45b5-bb89-882dfdb2be4b

Response

Code: 200 (Success)

```
{
  "resourceType": "Bundle",
  "id": "ed4425dd-7479-4b52-b404-502697281ee0",
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Procedure?_count=10&patient=17e77411359-8837d878-ea1e-45b5-bb89-882dfdb2be4b&_page=1"
    }
  ],
  "entry": [
    {
      "id": "17e7ba24dee-bf67ef6b-6419-4c7c-b905-a107b26bf8fb",
      "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Procedure/17e7ba24dee-bf67ef6b-6419-4c7c-b905-a107b26bf8fb",
      "resource": {
        "resourceType": "Procedure",
        "id": "17e7ba24dee-bf67ef6b-6419-4c7c-b905-a107b26bf8fb",
        "meta": {
          "versionId": "1",
          "lastUpdated": "2022-01-21T07:55:41.422Z",
          "profile": [
            "http://hl7.org/fhir/us/core/StructureDefinition/us-core-procedure"
          ]
        },
        "status": "completed",
        "code": {
          "coding": [
            {
              "system": "http://snomed.info/sct",
              "code": "430193006",
              "display": "Medication Reconciliation (procedure)"
            }
          ]
        },
        "text": "Medication Reconciliation (procedure)"
      },
      "subject": {
        "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882dfdb2be4b"
      },
      "encounter": {
        "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
      },
      "performedDateTime": "1941-02-13T18:33:18-05:00",
      "location": {
        "reference": "Location/17e7750e68f-6266aec8-0cd3-443e-adb9-be37f3bf25f2",
        "display": "PCP170967"
      }
    },
    {
      "search": {
        "mode": "match",
        "score": 1
      }
    }
  ]
}
```

Request: Provenance

GET https://[Base_URL]/Provenance/[id]

e.g.

https://[Base_URL]/Provenance/17e8adbdd12-21675728-3f52-4320-8b3b-d5977915336c

Response

Code: 200 (Success)

```
{
  "resourceType": "Provenance",
  "id": "17e8adbdd12-21675728-3f52-4320-8b3b-d5977915336c",
  "meta": {
    "versionId": "5",
    "lastUpdated": "2022-01-31T12:51:12.934Z",
    "profile": [
      "http://hl7.org/fhir/us/core/StructureDefinition/us-core-provenance"
    ]
  },
  "target": [
    {
      "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b"
    },
    {
      "reference": "Location/17e7750e68f-6266aec8-0cd3-443e-adb9-be37f3bf25f2"
    },
    {
      "reference": "Organization/17e774dca91-e7e4e9e3-390e-4c10-83ae-88beabc176e2"
    },
    {
      "reference": "Practitioner/17e774bb850-6b7b7f50-ed52-4523-a3a8-60fb0ef71c5b"
    },
    {
      "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
    },
    {
      "reference": "AllergyIntolerance/17e7758c319-7955ecfe-94da-4071-8488-4ef21425ed85"
    },
    {
      "reference": "CareTeam/17e775e7546-36e6a4cd-264e-4a94-92fa-cdf1b2fa30c1"
    },
    {
      "reference": "CarePlan/17e776e13ff-1ea7505d-f475-4f49-adde-30e69df86734"
    },
    {
      "reference": "Condition/17e7777ce8c-c66752fb-ee1b-40c2-ba5a-2e6ee89b3e1a"
    },
    {
      "reference": "Condition/17eb029f925-40603327-0ac3-48dd-972b-6c882b8751f5"
    },
    {
      "reference": "Device/17e777b3f34-04bc3627-92d3-43bb-92e7-35533110f750"
    },
    {
      "reference": "Observation/17e77809e90-d5cfd26d-ab5d-4f36-bc32-1ee799b2c278"
    },
    {
      "reference": "Observation/17e7785ea69-6c9f88d9-0553-4216-be17-f908ebe0b371"
    },
    {
      "reference": "Observation/17e778a9113-66c19e50-1b3f-41fd-a4da-ef86393fb0b2"
    },
    {
      "reference": "Observation/17e778d42db-92528d6f-7103-40fb-99ca-b2ac8837210d"
    }
  ]
}
```

```
"reference": "Observation/17e7791dd19-11eaf5c1-a5c0-429c-9867-f3110c743572"
},
{
  "reference": "Observation/17e77947034-7677a2e6-33af-4e06-83ae-a6727ac79885"
},
{
  "reference": "Observation/17e7797685b-e707227d-0cb5-4b28-bab2-ca1f4a1ba111"
},
{
  "reference": "Observation/17e7799a564-8f9ed437-d396-46bf-8f68-bf728955654c"
},
{
  "reference": "Observation/17e779bc672-37c76705-1f73-4531-bfcb-ee3cd3e25bfd"
},
{
  "reference": "Observation/17e77a065f3-77947ea0-1b53-47a5-9a9e-10f6e169f292"
},
{
  "reference": "Observation/17e77a29d41-85ec6814-a99a-4a73-ae02-f6a494a11468"
},
{
  "reference": "Observation/17e77a53c30-b356fe0d-d90b-42ed-91f9-5ed08c8047ca"
},
{
  "reference": "Observation/17e7c606d8a-9aeab44c-0261-465d-8930-4a218b201bf5"
},
{
  "reference": "DiagnosticReport/17e7b6e6f1c-119e9b8b-11b9-448b-b760-dc116013b654"
},
{
  "reference": "DiagnosticReport/17e7b741f29-4b6b813d-d11e-4597-8178-5bc5996ac4e4"
},
{
  "reference": "DiagnosticReport/17e7b7b7e66-36f4cf84-3482-4e36-8cca-13cc832b3bed"
},
{
  "reference": "DiagnosticReport/17e7d0e6d1d-c0e0136b-81d4-4164-a8b8-863694276a4f"
},
{
  "reference": "DocumentReference/17e7b838c61-4be99b80-0f94-4b26-a086-a30ed78c91ce"
},
{
  "reference": "DocumentReference/17e7d144d28-7d79ed3e-ca9e-400d-a9ed-add23f5cef07"
},
{
  "reference": "DocumentReference/17e7d19866c-dbbaa2e9-4d81-44bb-af5d-ec2e829e9c72"
},
{
  "reference": "DocumentReference/17e7d1c49cd-bfb30b25-9f98-40c7-9d94-977b7b803f21"
},
{
  "reference": "DocumentReference/17e7d24c947-205cd8e5-3a93-4ab9-8662-5e92b0080af7"
},
{
  "reference": "Goal/17e7b8b62f8-a4f77df6-b97e-4ece-82ff-36480c9cc38c"
},
{
  "reference": "Immunization/17e7b8dc9ab-8abc85e3-0ab1-470f-ae8-1f4e9a3597bb"
},
{
  "reference": "Immunization/17eb02c0f47-3b8a6c05-ae47-4002-9001-fd61626bfca5"
},
{
  "reference": "Medication/17e7b8f47d0-2bf2943a-123e-4725-b5dd-f5200df6c3a2"
},
{
```



```

"reference": "MedicationRequest/17e7b94dd8d-d3b0e43a-de55-4822-8bb5-07e5fd6177d9"
},
{
  "reference": "MedicationRequest/17e7b990c43-4525f6a4-81a0-40f7-97c7-7dbebff6e34b"
},
{
  "reference": "MedicationRequest/17e7b9b263e-9cd8f64f-f7e1-447e-8583-a753342d13c6"
},
{
  "reference": "MedicationRequest/17e7ba05cb1-0a5defbd-deb9-4792-9952-8f7f0e5bc74b"
},
{
  "reference": "Procedure/17e7ba24dee-bf67ef6b-6419-4c7c-b905-a107b26bf8fb"
},
{
  "reference": "Observation/17e81e97c4b-ea761488-0b67-43d0-a0d7-346dea549bee"
},
{
  "reference": "Observation/17e81ebac04-75d9f0f2-fcb2-4e2f-9c22-87fb7ddf5f5e"
},
{
  "reference": "Observation/17e81ed9c56-bcda0427-df92-4b58-b3ae-ac11c22f4d7d"
},
{
  "reference": "Observation/17e81ef5b5e-40df3e7d-19bc-4c2a-94bc-9a04ff93537c"
},
{
  "reference": "Observation/17e81f0fd34-6fc95b32-8346-4aec-9de3-072d12c992a5"
},
{
  "reference": "Observation/17e81f291e7-c99c91ac-57dd-47f6-b7f7-a365d2c087d7"
},
{
  "reference": "Observation/17e81f4a61d-ff3926f2-e35b-466a-bb6c-62a061425258"
},
{
  "reference": "Observation/17e8501692e-8c2d145f-fb69-4ae3-85d1-ef31ebf4a6a2"
},
{
  "reference": "Observation/17e8502f180-3c7e4770-3630-4654-922a-3f802c2939b1"
}
],
"recorded": "2020-12-10T18:33:18.715-05:00",
"agent": [
  {
    "type": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/provenance-participant-type",
          "code": "author",
          "display": "Author"
        }
      ],
      "text": "Author"
    },
    "who": {
      "reference": "Practitioner/17e774bb850-6b7b7f50-ed52-4523-a3a8-60fb0ef71c5b",
      "display": "Dr. Blossom971 Christiansen251"
    },
    "onBehalfOf": {
      "reference": "Organization/17e774dca91-e7e4e9e3-390e-4c10-83ae-88beabc176e2",
      "display": "PCP170967"
    }
  },
  {
    "type": {

```

```
"coding": [
  {
    "system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-provenance-participant-type",
    "code": "transmitter",
    "display": "Transmitter"
  }
],
"text": "Transmitter"
},
"who": {
  "reference": "Practitioner/17e774bb850-6b7b7f50-ed52-4523-a3a8-60fb0ef71c5b",
  "display": "Dr. Blossom971 Christiansen251"
},
"onBehalfOf": {
  "reference": "Organization/17e774dca91-e7e4e9e3-390e-4c10-83ae-88beabc176e2",
  "display": "PCP170967"
}
}
]
```

Request: Vital Signs

patient + code: GET [https://\[Base_URL\]/Observation?code=85354-9&patient=\[id\]](https://[Base_URL]/Observation?code=85354-9&patient=[id])

patient + category + date : GET [https://\[Base_URL\]/Observation?category=vital-signs&date=\[leYYYY-MM-DD\]&patient=\[id\]](https://[Base_URL]/Observation?category=vital-signs&date=[leYYYY-MM-DD]&patient=[id])

e.g.

[https://\[Base_URL\]/Observation?code=85354-9&patient=Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b](https://[Base_URL]/Observation?code=85354-9&patient=Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b)

[https://\[Base_URL\]/Observation?code=85354-9&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b](https://[Base_URL]/Observation?code=85354-9&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b)

[https://\[Base_URL\]/Observation?category=vital-signs&date=gt1994-05-18T19:33:18-04:00&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b](https://[Base_URL]/Observation?category=vital-signs&date=gt1994-05-18T19:33:18-04:00&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b)

Response

Code: 200 (Success)

```
{
  "resourceType": "Bundle",
  "id": "231df7ad-8e46-4371-99cc-4ce0dc9c5cb0",
  "type": "searchset",
  "total": 6,
  "link": [
    {
      "relation": "self",
      "url": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Observation?_count=10&category=vital-signs&date=gt1994-05-18T19:33:18-04:00&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b&_page=1"
    }
  ],
  "entry": [
    {
      "id": "17e7797685b-e707227d-0cb5-4b28-bab2-ca1f4a1ba111",
      "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Observation/17e7797685b-e707227d-0cb5-4b28-bab2-ca1f4a1ba111",
      "resource": {
        "resourceType": "Observation",
        "id": "17e7797685b-e707227d-0cb5-4b28-bab2-ca1f4a1ba111",
        "meta": {
          "versionId": "1",
          "lastUpdated": "2022-01-20T13:05:18.427Z",
          "profile": [
            "http://hl7.org/fhir/StructureDefinition/bodytemp",
            "http://hl7.org/fhir/StructureDefinition/vitalsigns"
          ]
        },
        "status": "final",
        "category": [
          {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/observation-category",
                "code": "vital-signs",
                "display": "vital-signs"
              }
            ]
          }
        ],
        "code": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "8310-5",
              "display": "Body temperature"
            },
            {
              "system": "http://loinc.org",
              "code": "8331-1",
              "display": "Oral temperature"
            }
          ]
        }
      }
    }
  ],
}
```

```

    "text": "Body temperature"
  },
  "subject": {
    "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b"
  },
  "encounter": {
    "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
  },
  "effectiveDateTime": "2009-08-19T19:33:18-04:00",
  "issued": "2009-08-19T19:33:18.715-04:00",
  "valueQuantity": {
    "value": 37.799,
    "unit": "Cel",
    "system": "http://unitsofmeasure.org",
    "code": "Cel"
  }
},
"search": {
  "mode": "match",
  "score": 1
}
},
{
  "id": "17e7799a564-8f9ed437-d396-46bf-8f68-bf728955654c",
  "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Observation/17e7799a564-8f9ed437-d396-46bf-8f68-bf728955654c",
  "resource": {
    "resourceType": "Observation",
    "id": "17e7799a564-8f9ed437-d396-46bf-8f68-bf728955654c",
    "meta": {
      "versionId": "1",
      "lastUpdated": "2022-01-20T13:07:45.124Z",
      "profile": [
        "http://hl7.org/fhir/StructureDefinition/bp",
        "http://hl7.org/fhir/StructureDefinition/vitalsigns"
      ]
    },
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "vital-signs",
            "display": "vital-signs"
          }
        ]
      }
    ]
  },
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "85354-9",
        "display": "Blood Pressure"
      }
    ]
  },
  "text": "Blood Pressure"
},
"subject": {
  "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b"
},
"encounter": {
  "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
},
"effectiveDateTime": "1994-05-19T19:33:18-04:00",
"issued": "1994-05-19T19:33:18.715-04:00",

```

```

"component": [
  {
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "8462-4",
          "display": "Diastolic Blood Pressure"
        }
      ],
      "text": "Diastolic Blood Pressure"
    },
    "valueQuantity": {
      "value": 85,
      "unit": "mm[Hg]",
      "system": "http://unitsofmeasure.org",
      "code": "mm[Hg]"
    }
  },
  {
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "8480-6",
          "display": "Systolic Blood Pressure"
        }
      ],
      "text": "Systolic Blood Pressure"
    },
    "valueQuantity": {
      "value": 129,
      "unit": "mm[Hg]",
      "system": "http://unitsofmeasure.org",
      "code": "mm[Hg]"
    }
  }
],
"search": {
  "mode": "match",
  "score": 1
},
{
  "id": "17e77a29d41-85ec6814-a99a-4a73-ae02-f6a494a11468",
  "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Observation/17e77a29d41-85ec6814-a99a-4a73-ae02-f6a494a11468",
  "resource": {
    "resourceType": "Observation",
    "id": "17e77a29d41-85ec6814-a99a-4a73-ae02-f6a494a11468",
    "meta": {
      "versionId": "1",
      "lastUpdated": "2022-01-20T13:17:32.865Z",
      "profile": [
        "http://hl7.org/fhir/StructureDefinition/hearttrate",
        "http://hl7.org/fhir/StructureDefinition/vitalsigns"
      ]
    },
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "vital-signs",
            "display": "vital-signs"
          }
        ]
      }
    ]
  }
}

```

```

    }
  ]
}
],
"code": {
  "coding": [
    {
      "system": "http://loinc.org",
      "code": "8867-4",
      "display": "Heart rate"
    }
  ],
  "text": "Heart rate"
},
"subject": {
  "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882dfb2be4b"
},
"encounter": {
  "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
},
"effectiveDateTime": "2003-07-10T19:33:18-04:00",
"issued": "2003-07-10T19:33:18.715-04:00",
"valueQuantity": {
  "value": 65,
  "unit": "/min",
  "system": "http://unitsofmeasure.org",
  "code": "/min"
}
},
"search": {
  "mode": "match",
  "score": 1
}
},
{
  "id": "17e81ef5b5e-40df3e7d-19bc-4c2a-94bc-9a04ff93537c",
  "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Observation/17e81ef5b5e-40df3e7d-19bc-4c2a-94bc-9a04ff93537c",
  "resource": {
    "resourceType": "Observation",
    "id": "17e81ef5b5e-40df3e7d-19bc-4c2a-94bc-9a04ff93537c",
    "meta": {
      "versionId": "1",
      "lastUpdated": "2022-01-22T13:17:34.43Z",
      "source": "#iLFSV7OLv0KF8dmQ",
      "profile": [
        "http://hl7.org/fhir/StructureDefinition/vitalsigns",
        "http://hl7.org/fhir/StructureDefinition/bodytemp"
      ]
    },
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "vital-signs",
            "display": "vital-signs"
          }
        ]
      }
    ]
  },
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "8310-5",

```

```

      "display": "Body temperature"
    },
    {
      "system": "http://loinc.org",
      "code": "8331-1",
      "display": "Oral temperature"
    }
  ],
  "text": "Body temperature"
},
"subject": {
  "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b"
},
"encounter": {
  "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
},
"effectiveDateTime": "2009-08-19T19:33:18-04:00",
"issued": "2009-08-19T19:33:18.715-04:00",
"dataAbsentReason": {
  "coding": [
    {
      "system": "http://terminology.hl7.org/CodeSystem/data-absent-reason",
      "code": "unknown",
      "display": "Unknown"
    }
  ],
  "text": "Unknown"
}
},
"search": {
  "mode": "match",
  "score": 1
}
},
{
  "id": "17e81f0fd34-6fc95b32-8346-4aec-9de3-072d12c992a5",
  "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Observation/17e81f0fd34-6fc95b32-8346-4aec-9de3-072d12c992a5",
  "resource": {
    "resourceType": "Observation",
    "id": "17e81f0fd34-6fc95b32-8346-4aec-9de3-072d12c992a5",
    "meta": {
      "versionId": "1",
      "lastUpdated": "2022-01-22T13:19:21.396Z",
      "source": "#iLFSV7OLv0KF8dmQ",
      "profile": [
        "http://hl7.org/fhir/StructureDefinition/vitalsigns",
        "http://hl7.org/fhir/StructureDefinition/bp"
      ]
    },
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "vital-signs",
            "display": "vital-signs"
          }
        ]
      }
    ]
  },
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "85354-9",

```

```

      "display": "Blood Pressure"
    }
  ],
  "text": "Blood Pressure"
},
"subject": {
  "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b"
},
"encounter": {
  "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
},
"effectiveDateTime": "1994-05-19T19:33:18-04:00",
"issued": "1994-05-19T19:33:18.715-04:00",
"dataAbsentReason": {
  "coding": [
    {
      "system": "http://terminology.hl7.org/CodeSystem/data-absent-reason",
      "code": "unknown",
      "display": "Unknown"
    }
  ],
  "text": "Unknown"
},
"component": [
  {
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "8462-4",
          "display": "Diastolic Blood Pressure"
        }
      ],
      "text": "Diastolic Blood Pressure"
    },
    "dataAbsentReason": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/data-absent-reason",
          "code": "unknown",
          "display": "Unknown"
        }
      ],
      "text": "Unknown"
    }
  },
  {
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "8480-6",
          "display": "Systolic Blood Pressure"
        }
      ],
      "text": "Systolic Blood Pressure"
    },
    "dataAbsentReason": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/data-absent-reason",
          "code": "unknown",
          "display": "Unknown"
        }
      ],
      "text": "Unknown"
    }
  }
]

```



```

    }
  }
]
},
"search": {
  "mode": "match",
  "score": 1
}
},
{
  "id": "17e8502f180-3c7e4770-3630-4654-922a-3f802c2939b1",
  "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Observation/17e8502f180-3c7e4770-3630-4654-922a-3f802c2939b1",
  "resource": {
    "resourceType": "Observation",
    "id": "17e8502f180-3c7e4770-3630-4654-922a-3f802c2939b1",
    "meta": {
      "versionId": "1",
      "lastUpdated": "2022-01-23T03:37:49.696Z",
      "source": "#iLFSV7OLv0KF8dmQ",
      "profile": [
        "http://hl7.org/fhir/StructureDefinition/vitalsigns",
        "http://hl7.org/fhir/StructureDefinition/hearttrate"
      ]
    },
    "status": "final",
    "category": [
      {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/observation-category",
            "code": "vital-signs",
            "display": "vital-signs"
          }
        ]
      }
    ],
    "code": {
      "coding": [
        {
          "system": "http://loinc.org",
          "code": "8867-4",
          "display": "Heart rate"
        }
      ]
    },
    "text": "Heart rate"
  },
  "subject": {
    "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882dfb2be4b"
  },
  "encounter": {
    "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
  },
  "effectiveDateTime": "2003-07-10T19:33:18-04:00",
  "issued": "2003-07-10T19:33:18.715-04:00",
  "dataAbsentReason": {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/data-absent-reason",
        "code": "unknown",
        "display": "Unknown"
      }
    ],
    "text": "Unknown"
  }
},
"search": {

```

```
    "mode": "match",
    "score": 1
  }
}
]
```

Request: Organization

GET [https://\[Base_URL\]/Organization/\[id\]](https://[Base_URL]/Organization/[id])

e.g. [https://\[Base_URL\]/Organization/1811435a7ea-7a1b883e-e0f0-4cb2-b938-4b6fd8d60f0c](https://[Base_URL]/Organization/1811435a7ea-7a1b883e-e0f0-4cb2-b938-4b6fd8d60f0c)

Response

The resource returned is expected to conform to the US Core Organization Profile

Code: 200 (Success)

```
{
  "resourceType": "Organization",
  "id": "1811435a7ea-7a1b883e-e0f0-4cb2-b938-4b6fd8d60f0c",
  "meta": {
    "versionId": "2",
    "lastUpdated": "2024-11-19T12:34:19.059+05:30",
    "source": "#OtUNHgljdSif9gwC",
    "profile": [
      "http://hl7.org/fhir/us/core/StructureDefinition/us-core-organization"
    ]
  },
  "identifier": [
    {
      "system": "https://github.com/synthetichealth/synthea",
      "value": "c7459800-e42b-3144-a934-59585c9eacff"
    },
    {
      "system": "http://hl7.org/fhir/sid/us-npi",
      "value": "1144221847"
    }
  ],
  "active": true,
  "type": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/organization-type",
          "code": "prov",
          "display": "Healthcare Provider"
        }
      ],
      "text": "Healthcare Provider"
    }
  ],
  "name": "PCP170967",
  "telecom": [
    {
      "system": "phone",
      "value": "413-221-0546"
    }
  ],
  "address": [
    {
      "line": [
        "1111 ELM ST"
      ]
    }
  ],
}
```

```

    "city": "WEST SPRINGFIELD",
    "state": "MA",
    "postalCode": "01089-1540",
    "country": "US"
  }
}
}

```

Request: DocumentReference

GET [https://\[Base_URL\]/DocumentReference/\[id\]](https://[Base_URL]/DocumentReference/[id])

e.g. [https://\[Base_URL\]/DocumentReference/1811435a7ea-7a1b883e-e0f0-4cb2-b938-4b6fd8d60f0c](https://[Base_URL]/DocumentReference/1811435a7ea-7a1b883e-e0f0-4cb2-b938-4b6fd8d60f0c)

e.g. [https://\[Base_URL\]/DocumentReference?patient=id](https://[Base_URL]/DocumentReference?patient=id)

Response

The resource returned is expected to conform to the US Core DocumentReference Profile

Code: 200 (Success)

```

{
  "resourceType": "Bundle",
  "id": "09f9423b-44a7-49c0-87f7-92737aef20c5",
  "meta": {
    "lastUpdated": "2025-11-01T17:02:26.375+05:30"
  },
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/DocumentReference?_id=1811498a297-eb9f1a18-fcac-4b00-8300-3c94af0882fe"
    }
  ],
  "entry": [
    {
      "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/DocumentReference/1811498a297-eb9f1a18-fcac-4b00-8300-3c94af0882fe",
      "resource": {
        "resourceType": "DocumentReference",
        "id": "1811498a297-eb9f1a18-fcac-4b00-8300-3c94af0882fe",
        "meta": {
          "versionId": "2",
          "lastUpdated": "2025-10-31T19:24:59.701+05:30",
          "source": "#CTLPNVIMT2G1aloz",
          "profile": [
            "http://hl7.org/fhir/us/core/StructureDefinition/us-core-documentreference"
          ]
        },
        "identifier": [
          {
            "system": "urn:ietf:rfc:3986",
            "value": "urn:uuid:18114577bc3-0a8379b4-a2f5-4cf6-b637-3301e3c0b523"
          }
        ],
        "status": "superseded",
        "type": {
          "coding": [
            {
              "system": "http://loinc.org",
              "code": "11506-3",
              "display": "Progress note"
            }
          ]
        }
      }
    }
  ]
}

```

```
[{"text": "Progress note", "category": [{"coding": [{"system": "http://hl7.org/fhir/us/core/CodeSystem/us-core-documentreference-category", "code": "clinical-note", "display": "Clinical Note"}]}], "subject": {"reference": "Patient/18114240088-12a390c4-2550-4130-a7d2-ff4f30bb24d0"}, "date": "1979-02-22T18:34:18.715-05:00", "author": [{"reference": "Practitioner/18114325cd4-68093d66-195e-4b9e-8899-44bab4231e86", "display": "Dr. Blossom971 Christiansen251"}], "custodian": {"reference": "Organization/1811435a7ea-7a1b883e-e0f0-4cb2-b938-4b6fd8d60f0c", "display": "PCP170967"}, "content": [{"attachment": {"contentType": "text/plain", "data": "CjE5NzktMDItMjIwZDQyZWYyZWVudCBpbWVudAotIEVnbmZ1c2lrbG9tIE5hc2FsiENvbmduc3Rpb24KL SBTbmVlemluZyBGaXRzCi0gTmFzYWwgRGlzY2hhcmdlCi0gZDQyZWYyZWVudCBpbWVudAotIEVnbmZ1c2lrbG9tIE5hc2FsiENvbmduc3Rpb24KL hpc3RvcnkKUGF0aWVudCBpcyBtYXJyaWVkaLiBQYXRpZW50IGlziHRlbXBvcnFyaWx5IGhvbnVWVsZXNzLiBQYXRpZW50IGlziGFuLGFjdGl2ZSBzbW9rZXI gYW5kIGlziGFuLGFuZS9ob2xpYy4gUGF0aWVudCBpZGVudGlmaWVzIGFzIGhldGVyb3NieHVhbC4KICBhdGllbnQgY29tZXMGZnJvbSBhIG1pZGRsZSB zb2Npb2Vjb25vbWljIGJhY2tncm91bmQuIFBhdGllbnQgZGkiG5vdCBmaW5pc2ggaGlnaCBzY2hvbWUuIFBhdGllbnQgY3VyYmVudGVudGx5IGhlcycBBZXR uYS4KiMgQWxsZSJnaWVzCmxhdGV4IGFsbgGVyZ3ksIGFsbgGVyZ3kgdG8gc295YQoKIyBNZWRpY2F0aW9ucwpHY2V0YW1pbm9waGVuIDMyNSBtZ yBvcnFsiHRHrYmxldDsgMTlwIGFjdHVhdCBmbHV0aWNhc29uZSBwc9waW9uYXRlIDAuMDQ0IG1nL2FjdHVhdCBtZXRIcmVklGRvc2UgaW5oYWxl cjsbmrRhMDIwNTAzIDlwMCBhY3R1YXQgYWxidXRlcm9sIDAuMDkqbWcvYWN0dWF0IG1ldGVyZWQgZG9ZSBpbmhbbGVyOyBwcmVkbmlzb25lI DUgbWcg3JhbCB0YWJsZXQ7IGFzcGlyaW4gODEgbWcg3JhbCB0YWJsZXQKCiMgQXNzZXNzbWVudCBhbmQgUGxhbgoKCiMjIjFBsYW4KCIRoZSB mb2xsb3dpbmcmcHJyY2VkdXIjcyB3ZXJIIGNvbmr1Y3RIZDoKLSBtZWRpY2F0aW9uIHJlY29uY2lsaWF0aW9uChwcm9jZWRR1cmUpClRoZSBwYXRpZ W50IHdhcyBwcmVzY3pYmVklHROZSBmb2xsb3dpbmcmcHwVkaWNhdGlvbnM6Ci0gMTlwIGFjdHVhdCBmbHV0aWNhc29uZSBwc9waW9uYXRlI DAuMDQ0IG1nL2FjdHVhdCBtZXRIcmVklGRvc2UgaW5oYWxlclgtIG5kYTAYMDUwMyAyMDAgYWN0dWF0IGFsYnV0ZSVjbCAwLjA5IG1nL2FjdHVh dCBtZXRIcmVklGRvc2UgaW5oYWxlclgo="}}], "format": {"system": "http://ihe.net/fhir/ValueSet/IHE.FormatCode.codesystem", "code": "urn:ihe:iti:xds:2017:mimeTypeSufficient", "display": "mimeType Sufficient"}}, {"context": {"encounter": [{"reference": "Encounter/1811439a512-5e32771c-2a1f-4e1d-a42d-db93aba8556d"}]}, "period": {"start": "1979-02-22T18:33:18-05:00",
```

```

        "end": "1979-02-22T18:48:18-05:00"
      }
    }
  },
  "search": {
    "mode": "match"
  }
}
]
}

```

Request: ServiceRequest

This is a FHIR resource that takes care of the following CCDA sections:

1. Reason For Referral
2. SDOH Intervention

GET [https://\[Base_URL\]/ServiceRequest/\[id\]](https://[Base_URL]/ServiceRequest/[id])

e.g. [https://\[Base_URL\]/ServiceRequest/1811435a7ea-7a1b883e-e0f0-4cb2-b938-4b6fd8d60f0c](https://[Base_URL]/ServiceRequest/1811435a7ea-7a1b883e-e0f0-4cb2-b938-4b6fd8d60f0c)

e.g. [https://\[Base_URL\]/ServiceRequest?patient=id](https://[Base_URL]/ServiceRequest?patient=id)

Response

The resource returned is expected to conform to the US Core ServiceRequest Profile

Code: 200 (Success)

```

{
  "resourceType": "Bundle",
  "id": "cc11ac6e-4c04-4545-a83f-4566b9160ada",
  "meta": {
    "lastUpdated": "2025-11-01T17:09:56.724+05:30"
  },
  "type": "searchset",
  "total": 1,
  "link": [
    {
      "relation": "self",
      "url": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/ServiceRequest?_id=18f0b6042c5-d10448a0-792e-45ac-900c-fb8b9504a256"
    }
  ],
  "entry": [
    {
      "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/ServiceRequest/18f0b6042c5-d10448a0-792e-45ac-900c-fb8b9504a256",
      "resource": {
        "resourceType": "ServiceRequest",
        "id": "18f0b6042c5-d10448a0-792e-45ac-900c-fb8b9504a256",
        "meta": {
          "versionId": "1",
          "lastUpdated": "2024-11-19T12:34:23.708+05:30",
          "source": "#OtUNHgljdSif9gwC",
          "profile": [
            "http://hl7.org/fhir/us/core/StructureDefinition/us-core-servicerequest"
          ]
        },
        "identifier": [
          {
            "system": "https://github.com/inferno-framework/us-core-test-kit",
            "value": "ba89c2af-ce2b-405d-8eff-793eabe1c38d"
          }
        ],
        "status": "completed",
        "intent": "order",
        "category": [
          {
            "coding": [

```

```

    {
      "system": "http://snomed.info/sct",
      "code": "387713003",
      "display": "Surgical procedure"
    }
  ],
  "text": "Surgical procedure"
}
],
"code": {
  "coding": [
    {
      "system": "http://snomed.info/sct",
      "code": "233169004",
      "display": "Automatic defibrillator procedure (procedure)"
    }
  ]
},
"subject": {
  "reference": "Patient/18114240088-12a390c4-2550-4130-a7d2-ff4f30bb24d0"
},
"encounter": {
  "reference": "Encounter/1811439a512-5e32771c-2a1f-4e1d-a42d-db93aba8556d"
},
"occurrencePeriod": {
  "start": "2021-11-20",
  "end": "2021-12-04"
},
"authoredOn": "2015-03-30",
"requester": {
  "reference": "Practitioner/18114325cd4-68093d66-195e-4b9e-8899-44bab4231e86"
},
"reasonCode": [
  {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "71908006",
        "display": "Ventricular fibrillation (disorder)"
      }
    ],
    "text": "Bradycardia"
  }
]
},
"search": {
  "mode": "match"
}
}
]
}

```

Request: Coverage

GET [https://\[Base_URL\]/Coverage/{id}](https://[Base_URL]/Coverage/{id})

e.g. [https://\[Base_URL\]/Coverage/1811435a7ea-7a1b883e-e0f0-4cb2-b938-4b6fd8d60f0c](https://[Base_URL]/Coverage/1811435a7ea-7a1b883e-e0f0-4cb2-b938-4b6fd8d60f0c)

e.g. [https://\[Base_URL\]/Coverage?patient=id](https://[Base_URL]/Coverage?patient=id)

Response

The resource returned is expected to conform to the US Core Coverage Profile

Code: 200 (Success)

```

{
  "resourceType": "Bundle",

```

```
"id": "f2ecd6e3-1826-4af9-8bef-beded3c61282",
"meta": {
  "lastUpdated": "2025-11-01T17:01:39.511+05:30"
},
"type": "searchset",
"total": 1,
"link": [
  {
    "relation": "self",
    "url": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Coverage/_search"
  }
],
"entry": [
  {
    "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Coverage/18f0a59193c-e99cbeb5-f2c3-406c-9f6d-a84ce586ff9a",
    "resource": {
      "resourceType": "Coverage",
      "id": "18f0a59193c-e99cbeb5-f2c3-406c-9f6d-a84ce586ff9a",
      "meta": {
        "versionId": "1",
        "lastUpdated": "2024-11-19T12:34:21.915+05:30",
        "source": "#OtUNHgljdSif9gWC",
        "profile": [
          "http://hl7.org/fhir/us/core/StructureDefinition/us-core-coverage"
        ]
      },
      "identifier": [
        {
          "type": {
            "coding": [
              {
                "system": "http://terminology.hl7.org/CodeSystem/v2-0203",
                "code": "MB"
              }
            ]
          },
          "system": "https://github.com/inferno-framework/us-core-test-kit",
          "value": "f4a375d2-4e53-4f81-ba95-345e7573b550"
        }
      ],
      "status": "active",
      "type": {
        "coding": [
          {
            "system": "https://nahdo.org/sopt",
            "code": "3712",
            "display": "PPO"
          },
          {
            "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
            "code": "PPO",
            "display": "preferred provider organization policy"
          }
        ]
      },
      "text": "PPO"
    },
    "subscriberId": "888009335",
    "beneficiary": {
      "reference": "Patient/18114240088-12a390c4-2550-4130-a7d2-ff4f30bb24d0"
    },
    "relationship": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/subscriber-relationship",
          "code": "self"
        }
      ]
    }
  }
]
```

```

    ],
    "text": "Self"
  },
  "period": {
    "start": "2020-01-01"
  },
  "payor": [
    {
      "reference": "Organization/1811435a7ea-7a1b883e-e0f0-4cb2-b938-4b6fd8d60f0c",
      "display": "PCP87052"
    }
  ],
  "class": [
    {
      "type": {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/coverage-class",
            "code": "group",
            "display": "Group"
          }
        ]
      },
      "text": "An employee group"
    },
    {
      "value": "xyz",
      "name": "XYZ employee Group Plan"
    }
  ],
  {
    "type": {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/coverage-class",
          "code": "plan",
          "display": "Plan"
        }
      ]
    },
    "text": "A specific suite of benefits."
  },
  {
    "value": "gold",
    "name": "Acme Gold Plan"
  }
]
},
"search": {
  "mode": "match"
}
}
]
}

```

Request: MedicationDispense

GET [https://\[Base_URL\]/MedicationDispense/\[id\]](https://[Base_URL]/MedicationDispense/[id])

e.g. [https://\[Base_URL\]/MedicationDispense/1811435a7ea-7a1b883e-e0f0-4cb2-b938-4b6fd8d60f0c](https://[Base_URL]/MedicationDispense/1811435a7ea-7a1b883e-e0f0-4cb2-b938-4b6fd8d60f0c)

e.g. [https://\[Base_URL\]/MedicationDispense?patient=id](https://[Base_URL]/MedicationDispense?patient=id)

Response

The resource returned is expected to conform to the US Core MedicationDispense Profile

Code: 200 (Success)

```

{
  "resourceType": "Bundle",
  "id": "0ef99c93-a771-48f6-a2a2-77fb3d0ef13a",
  "meta": {

```



```
"lastUpdated": "2025-11-01T17:02:56.829+05:30"
},
"type": "searchset",
"total": 1,
"link": [
  {
    "relation": "self",
    "url": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/MedicationDispense?patient=18114240088-12a390c4-2550-4130-a7d2-ff4f30bb24d0"
  }
],
"entry": [
  {
    "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/MedicationDispense/3f4923e1-109a-4af2-93e2-dd744b6dcd4c",
    "resource": {
      "resourceType": "MedicationDispense",
      "id": "3f4923e1-109a-4af2-93e2-dd744b6dcd4c",
      "meta": {
        "versionId": "3",
        "lastUpdated": "2024-12-09T11:08:54.204+05:30",
        "source": "#rnoQ8mqRJgZ20fqP",
        "profile": [
          "http://hl7.org/fhir/us/core/StructureDefinition/us-core-medicationdispense"
        ]
      },
      "identifier": [
        {
          "system": "https://github.com/inferno-framework/us-core-test-kit",
          "value": "1bc16d93-9ac8-4dc7-ad90-e649c01f8b10"
        }
      ],
      "status": "completed",
      "medicationReference": {
        "reference": "Medication/18114a0741e-2ab42c88-6cd0-49e2-8147-fb9954e1f4e1",
        "display": "10 ML Furosemide 10 MG/ML Injection"
      },
      "subject": {
        "reference": "Patient/18114240088-12a390c4-2550-4130-a7d2-ff4f30bb24d0"
      },
      "performer": [
        {
          "actor": {
            "reference": "Practitioner/18114325cd4-68093d66-195e-4b9e-8899-44bab4231e86",
            "display": "Dr. Willena258 Oberbrunner298"
          }
        }
      ],
      "authorizingPrescription": [
        {
          "reference": "MedicationRequest/18114a6a311-59175c8a-f101-4638-953c-82f783333830",
          "display": "10 ML Furosemide 10 MG/ML Injection"
        }
      ],
      "type": {
        "coding": [
          {
            "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
            "code": "FFC",
            "display": "First Fill - Complete"
          }
        ]
      },
      "quantity": {
        "value": 480,
        "unit": "mL",
        "system": "http://unitsofmeasure.org",

```

```

    "code": "mL"
  },
  "whenHandedOver": "2016-12-08T06:38:52Z",
  "dosageInstruction": [
    {
      "text": "10 mL bid",
      "timing": {
        "repeat": {
          "boundsPeriod": {
            "start": "2008-04-05"
          },
          "frequency": 2,
          "period": 1,
          "periodUnit": "d"
        }
      },
      "doseAndRate": [
        {
          "doseQuantity": {
            "value": 10,
            "unit": "mL",
            "system": "http://unitsofmeasure.org",
            "code": "mL"
          }
        }
      ]
    }
  ],
  "search": {
    "mode": "match"
  }
}

```

Request: Specimen

GET [https://\[Base_URL\]/Specimen/\[id\]](https://[Base_URL]/Specimen/[id])

e.g. [https://\[Base_URL\]/Specimen/1811435a7ea-7a1b883e-e0f0-4cb2-b938-4b6fd8d60f0c](https://[Base_URL]/Specimen/1811435a7ea-7a1b883e-e0f0-4cb2-b938-4b6fd8d60f0c)

Response

The resource returned is expected to conform to the US Core SpecimenProfile

Code: 200 (Success)

```

{
  "resourceType": "Specimen",
  "id": "18f0abe1999-9afa3deb-e111-4a8f-a8c3-914d100b49d8",
  "meta": {
    "versionId": "1",
    "lastUpdated": "2024-11-19T12:34:22.045+05:30",
    "source": "#OtUNHgljdSif9gwC",
    "profile": [
      "http://hl7.org/fhir/us/core/StructureDefinition/us-core-specimen"
    ]
  },
  "identifier": [
    {
      "system": "https://github.com/inferno-framework/us-core-test-kit",
      "value": "82115a91-04c9-4654-acb5-72f2806e00fc"
    }
  ],
  "accessionIdentifier": {
    "system": "https://github.com/inferno-framework/us-core-test-kit",

```

```

"value": "7446d969-3293-4f51-8d48-034121156b4e"
},
"type": {
  "coding": [
    {
      "system": "http://snomed.info/sct",
      "code": "122575003",
      "display": "Urine specimen (specimen)"
    }
  ]
},
"subject": {
  "reference": "Patient/18114240088-12a390c4-2550-4130-a7d2-ff4f30bb24d0"
},
"collection": {
  "collectedDateTime": "2005-07-05T18:07:12Z",
  "bodySite": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "106004",
        "display": "Posterior carpal region"
      }
    ]
  }
},
"container": [
  {
    "type": {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "702244006",
          "display": "Sterile urine specimen container"
        }
      ]
    },
    "text": "Sterile urine specimen container"
  }
],
"condition": [
  {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/v2-0493",
        "code": "AUT",
        "display": "Autolyzed"
      }
    ]
  }
]
}

```

Request: RelatedPerson

GET [https://\[Base_URL\]/RelatedPerson/\[id\]](https://[Base_URL]/RelatedPerson/[id])

e.g. [https://\[Base_URL\]/RelatedPerson/1811435a7ea-7a1b883e-e0f0-4cb2-b938-4b6fd8d60f0c](https://[Base_URL]/RelatedPerson/1811435a7ea-7a1b883e-e0f0-4cb2-b938-4b6fd8d60f0c)

Response

The resource returned is expected to conform to the US Core RelatedPerson Profile

Code: 200 (Success)

```

{
  "resourceType": "RelatedPerson",

```

```
"id": "re1a0ab5-5e30-46b2-868d-7e56c252b1f0",
"meta": {
  "versionId": "1",
  "lastUpdated": "2024-12-09T12:40:19.084+05:30",
  "source": "#sCPUceOmdiu7NGzZ",
  "profile": [
    "http://hl7.org/fhir/us/core/StructureDefinition/us-core-relatedperson"
  ],
},
"active": false,
"patient": {
  "reference": "Patient/18114240088-12a390c4-2550-4130-a7d2-ff4f30bb24d0"
},
"relationship": [
  {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/v3-RoleCode",
        "code": "ROOM",
        "display": "Roommate"
      }
    ]
  }
],
"name": [
  {
    "use": "official",
    "family": "Jefferson174",
    "given": [
      "Ronald408",
      "MacGyver246"
    ],
    "prefix": [
      "Mr."
    ]
  }
],
"telecom": [
  {
    "system": "phone",
    "value": "555-164-3435",
    "use": "home"
  }
],
"address": [
  {
    "extension": [
      {
        "url": "http://hl7.org/fhir/StructureDefinition/geolocation",
        "extension": [
          {
            "url": "latitude",
            "valueDecimal": 42.33734137767395
          },
          {
            "url": "longitude",
            "valueDecimal": -71.0633497363917
          }
        ]
      }
    ],
    "line": [
      "445 Braun Mews"
    ],
    "city": "Boston",
    "state": "MA",
  }
]
```

```

    "postalCode": "02110",
    "country": "US",
    "period": {
      "start": "2019-02-07"
    }
  }
}
]
}

```

Request: Observation

This is a FHIR resource that takes care of the following CCDA sections:

1. Pregnancy Status
2. Pregnancy Intent
3. Occupation
4. Respiratory Rate
5. Heart Rate
6. Body Temperature
7. Pediatric Weight For Height
8. Pulse Oximetry
9. Head Circumference
10. Body Height
11. BMI
12. Screening Assessment
13. Blood Pressure
14. Clinical Result
15. Pediatric BMI
16. Pediatric Head Occipital Frontal Circumference Percentile
17. Observation Body Weight
18. SDOH Assessment
19. Basic Occupation and Occupation Industry
20. Tribal Affiliation
21. Functional Status
22. Cognitive Status
23. Disability Status
24. Gender Identity
25. Sexual Orientation
26. Sex at Birth

In the below GET URL, you need to mention appropriate [code] and you will get the required bundle.

GET [https://\[Base_URL\]/Observation?code=\[code\]&patient=\[id\]](https://[Base_URL]/Observation?code=[code]&patient=[id])

e.g. [https://\[Base_URL\]/Observation?code=72166-2&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b](https://[Base_URL]/Observation?code=72166-2&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b)

Response

Code: 200 (Success)

```

{
  "resourceType": "Bundle",
  "id": "2ea85043-b450-40f1-8d2d-751af1c10abb",
  "type": "searchset",
  "total": 2,
  "link": [
    {
      "relation": "self",
      "url": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Observation?_count=10&code=72166-2&patient=17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b&_page=1"
    }
  ],
  "entry": [
    {
      "id": "17e77809e90-d5cfd26d-ab5d-4f36-bc32-1ee799b2c278",
      "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Observation/17e77809e90-d5cfd26d-ab5d-4f36-bc32-1ee799b2c278",
      "resource": {

```

```

"resourceType": "Observation",
"id": "17e77809e90-d5cfd26d-ab5d-4f36-bc32-1ee799b2c278",
"meta": {
  "versionId": "1",
  "lastUpdated": "2022-01-20T12:40:24.976Z",
  "profile": [
    "http://hl7.org/fhir/us/core/StructureDefinition/us-core-smokingstatus"
  ]
},
"status": "final",
"category": [
  {
    "coding": [
      {
        "system": "http://terminology.hl7.org/CodeSystem/observation-category",
        "code": "survey",
        "display": "survey"
      }
    ]
  }
],
"code": {
  "coding": [
    {
      "system": "http://loinc.org",
      "code": "72166-2",
      "display": "Tobacco smoking status NHIS"
    }
  ]
},
"text": "Tobacco smoking status NHIS"
},
"subject": {
  "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882dfb2be4b"
},
"encounter": {
  "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
},
"effectivePeriod": {
  "start": "1940-09-05T19:33:18-04:00",
  "end": "1940-09-05T20:33:18-04:00"
},
"issued": "1940-09-05T19:33:18.715-04:00",
"valueCodeableConcept": {
  "coding": [
    {
      "system": "http://snomed.info/sct",
      "code": "266919005",
      "display": "Never smoker"
    }
  ]
},
"text": "Never smoker"
}
},
"search": {
  "mode": "match",
  "score": 1
}
},
{
  "id": "17e7785ea69-6c9f88d9-0553-4216-be17-f908ebe0b371",
  "fullUrl": "https://qafhir.kantimehealth.net:9443/fhirserver/fhir/Observation/17e7785ea69-6c9f88d9-0553-4216-be17-f908ebe0b371",
  "resource": {
    "resourceType": "Observation",
    "id": "17e7785ea69-6c9f88d9-0553-4216-be17-f908ebe0b371",
    "meta": {
      "versionId": "1",

```

```

    "lastUpdated": "2022-01-20T12:46:12.073Z",
    "profile": [
      "http://hl7.org/fhir/us/core/StructureDefinition/us-core-smokingstatus"
    ]
  },
  "status": "final",
  "category": [
    {
      "coding": [
        {
          "system": "http://terminology.hl7.org/CodeSystem/observation-category",
          "code": "survey",
          "display": "survey"
        }
      ]
    }
  ],
  "code": {
    "coding": [
      {
        "system": "http://loinc.org",
        "code": "72166-2",
        "display": "Tobacco smoking status NHIS"
      }
    ],
    "text": "Tobacco smoking status NHIS"
  },
  "subject": {
    "reference": "Patient/17e77411359-8837d878-ea1e-45b5-bb89-882fdb2be4b"
  },
  "encounter": {
    "reference": "Encounter/17e77566282-0c9c236b-f0bd-45e0-9c61-80991a376967"
  },
  "effectivePeriod": {
    "start": "1940-09-05T19:33:18-04:00",
    "end": "1940-09-05T20:33:18-04:00"
  },
  "issued": "1940-09-05T19:33:18.715-04:00",
  "valueCodeableConcept": {
    "coding": [
      {
        "system": "http://snomed.info/sct",
        "code": "266919005",
        "display": "Never smoker"
      }
    ],
    "text": "Never smoker"
  }
},
"search": {
  "mode": "match",
  "score": 1
}
}
}

```

Profile audience and scope

This profile is intended to be used by developers of apps that need to access user identity information or other FHIR resources by requesting authorization from OAuth 2.0 compliant authorization servers.

The profile defines a method through which an app requests authorization to access a FHIR resource, and then uses that authorization to retrieve the resource. Synchronization of patient context is not addressed; for use cases that require context synchronization (e.g., learning

about when the in-context patient changes within an EHR session) In other words, if the patient chart is changed during the session, the application will not inherently be updated.

Security and Privacy Considerations

App Protection

The app is responsible for protecting itself from potential misbehaving or malicious values passed to its redirect URL (e.g., values injected with executable code, such as SQL) and for protecting authorization codes, access tokens, and refresh tokens from unauthorized access and use. The app developer must be aware of potential threats, such as malicious apps running on the same platform, counterfeit authorization servers, and counterfeit resource servers, and implement countermeasures to help protect both the app itself and any sensitive information it may hold. For background, see the [OAuth 2.0 Threat Model and Security Considerations](#).

Specific requirements are:

- Apps SHALL ensure that sensitive information (authentication secrets, authorization codes, tokens) is transmitted ONLY to authenticated servers, over TLS-secured channels. We support TLS 1.2. Ensure that TLS 1.2 is enabled on our web server's operating system by modifying registry settings
- Apps SHALL generate an unpredictable **state** parameter for each user session; SHALL include **state** with all authorization requests; and SHALL validate the **state** value for any request sent to its redirect URL.
- An app SHALL NOT execute untrusted user-supplied inputs as code.
- App SHALL NOT forward values passed back to its redirect URL to any other arbitrary or user-provided URL (a practice known as an "open redirector").
- An app SHALL NOT store bearer tokens in cookies that are transmitted as clear text.
- Apps SHOULD persist tokens and other sensitive data in app-specific storage locations only, and SHOULD NOT persist them in system-wide-discoverable locations.
- Secure issuance of refresh token: Native applications call our OAuth api to request tokens. Since all our endpoints are on https they are secure.
- We support PKCE

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1. TERMS ACCEPTANCE AND REPRESENTATION

- a. Accepting the Terms. These Terms of Use ("Terms") govern your access to and use of the KANTIMEHEALTH EHR API, documentation, services, etc. By accessing or using the KANTIMEHEALTH APIs, you agree to be bound by these Terms. "KANTIMEHEALTH EHR API" means the API provided by KANTIMEHEALTH to allow authorized access to query our Client(s) Electronic Health Record system. You represent and warrant that you are at least 18 years of age and that you possess the legal right and ability to agree to these Terms and to use the KANTIMEHEALTH APIs in accordance with these Terms.
- b. Entity Representation. If you are using the KANTIMEHEALTH API on behalf of a legal entity (i.e. a Clinical software services company), you represent that you have proper authority to act on behalf of and bind the entity to these Terms, and by accepting, you accept on behalf of the entity (and all references to "you" in the Terms refer to the entity).

2. REPRESENTATIONS AND RESPONSIBILITIES

- a. Compliance. You agree to be financially responsible for your use of the KANTIMEHEALTH EHR APIs and to comply with your responsibilities and obligations as stated in these Terms. You agree to comply at all times with all applicable laws, rules and regulations relating to the use of the KANTIMEHEALTH EMR APIs. You hereby grant KANTIMEHEALTH the right to monitor and periodically audit in a reasonable manner your use of the KANTIMEHEALTH APIs, your App and other activities related to your obligations under these Terms.
- b. Virus Warranty. You warrant that your Apps will not contain any viruses or other malicious computer instructions, devices, or techniques that can or were designed to threaten, infect, damage, disable, or shut down the KANTIMEHEALTH EHR APIs, any technology, software, solution, equipment or any computer system.

3. GENERAL

- a. Changes. KANTIMEHEALTH may, in its sole and absolute discretion, make changes, modifications or updates to the KANTIMEHEALTH EHR API (including without limitation changes to the capabilities and tech specs), without notice to you.
- b. Global Availability. KANTIMEHEALTH makes no representations that the KANTIMEHEALTH EHR APIs are appropriate or available for use in locations outside of the United States, and access to them from such territories is at your own risk. Those who choose to access the KANTIMEHEALTH APIs from locations outside of the United States do so at their own initiative and are responsible for compliance with applicable local laws.
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 - f. Term and Termination. Either you or KANTIMEHEALTH may terminate your right to use the KANTIMEHEALTH EHR APIs at any time, with or without cause, upon notice. KANTIMEHEALTH also reserves the right to disable your API access in a production environment at any time, with or without cause. KANTIMEHEALTH reserves the right to disable access to the KANTIMEHEALTH EHR APIs if your App poses any security, privacy, or patient safety risks. The provisions concerning Indemnification, Waiver, Release and Limitation of Liability, and General shall survive any termination of these Terms.
 - g. Governing Law. These Terms are governed by US federal law or the laws of the State of California
4. Application developer affirmations to Certified API Developers regarding the ability of their applications to secure a refresh token, a client secret, or both, must be treated in a good faith manner.